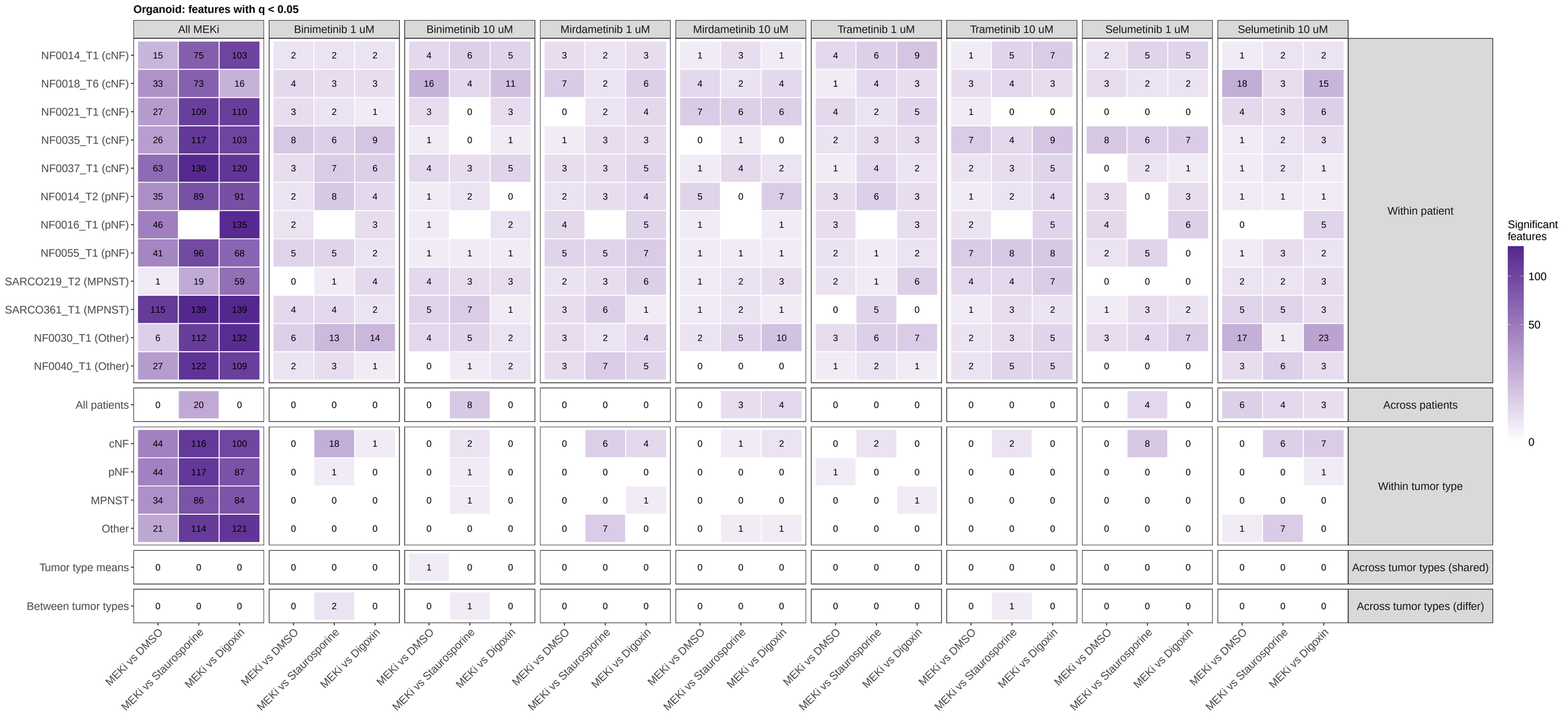
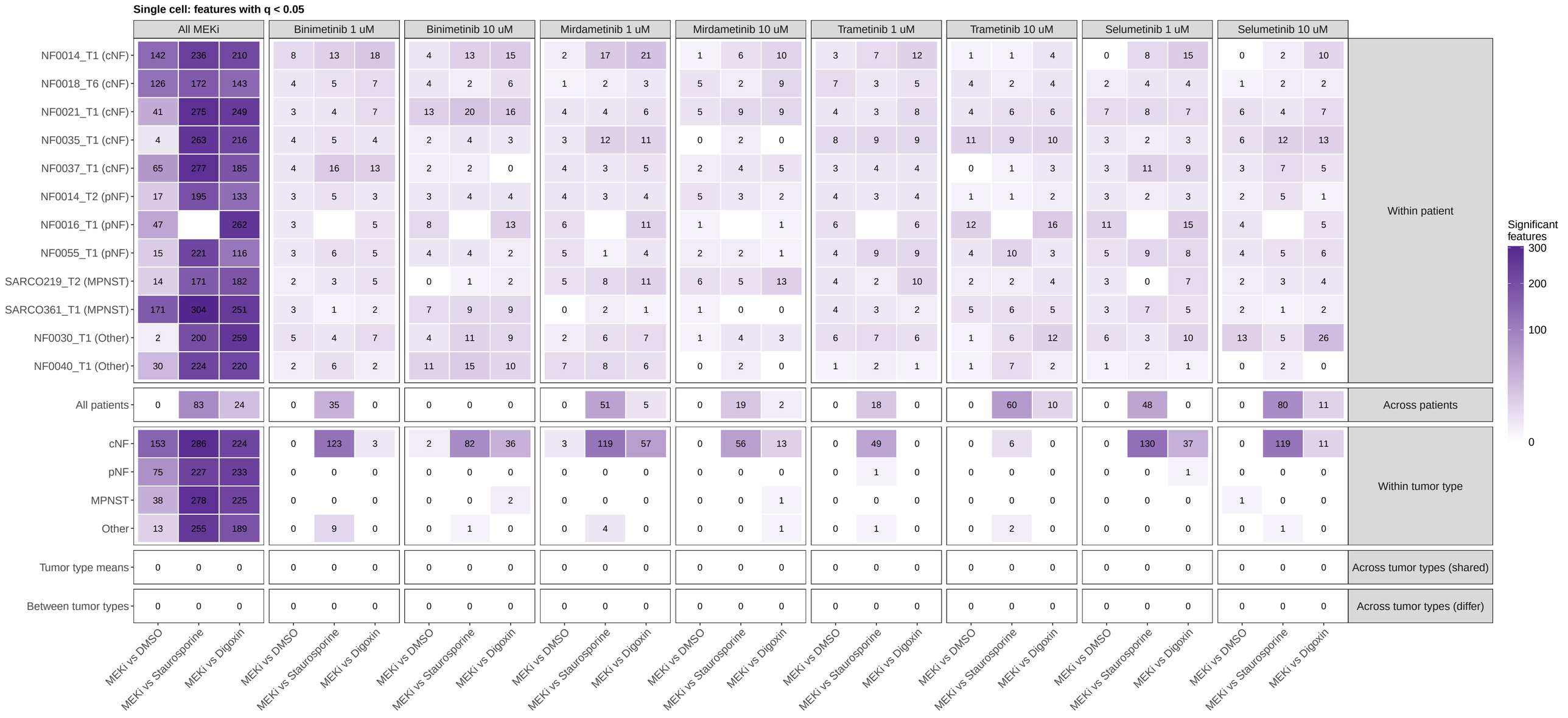
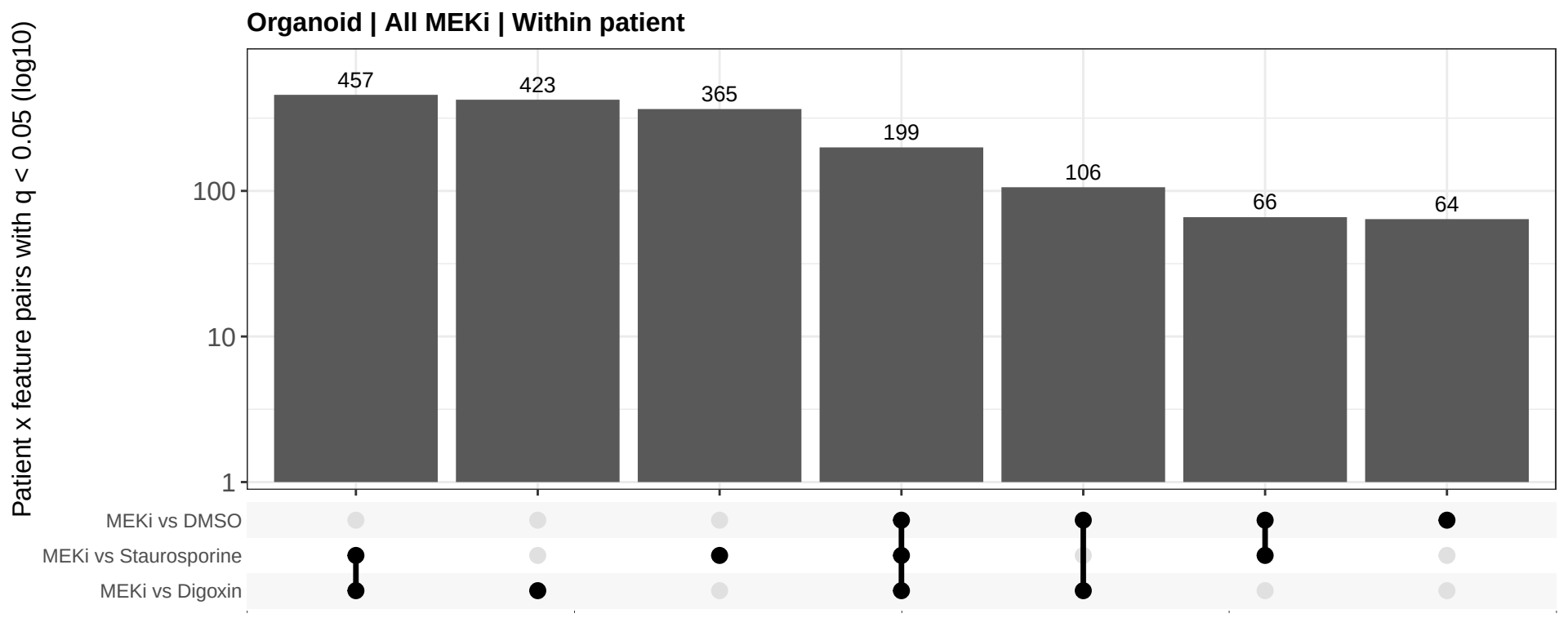
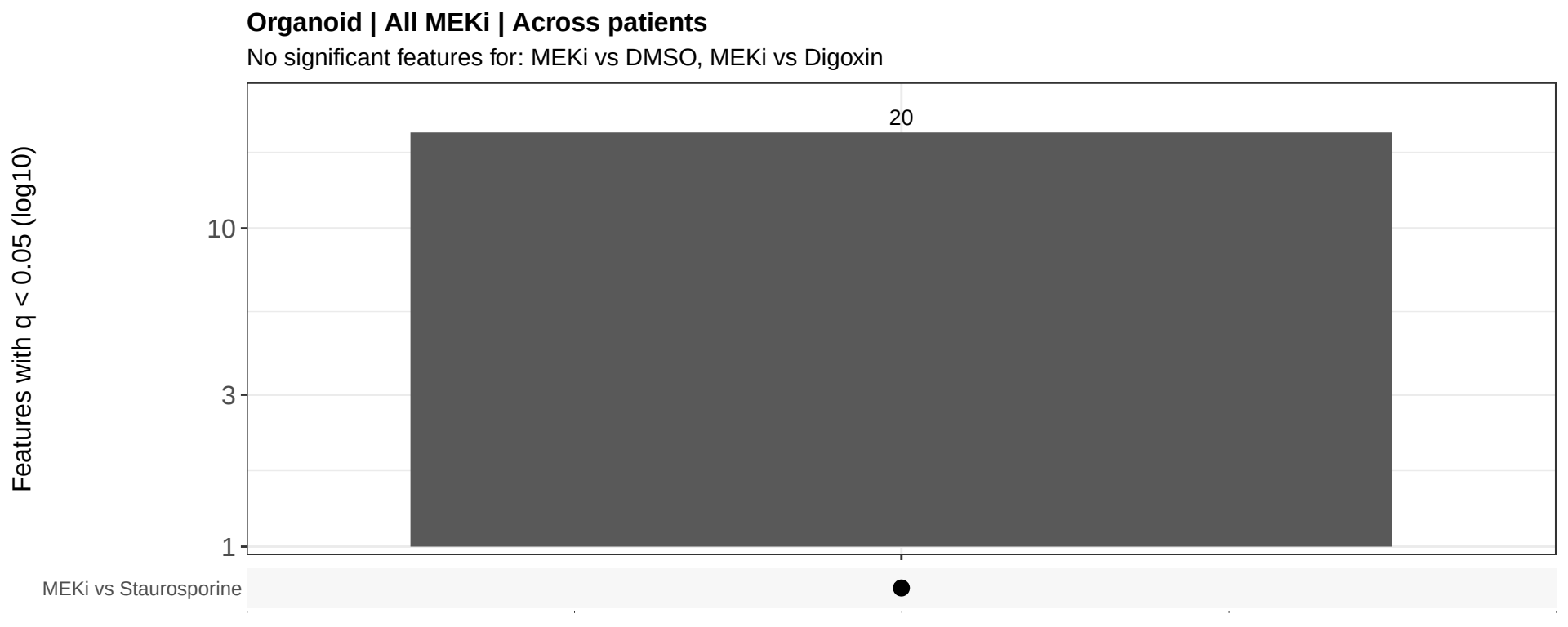
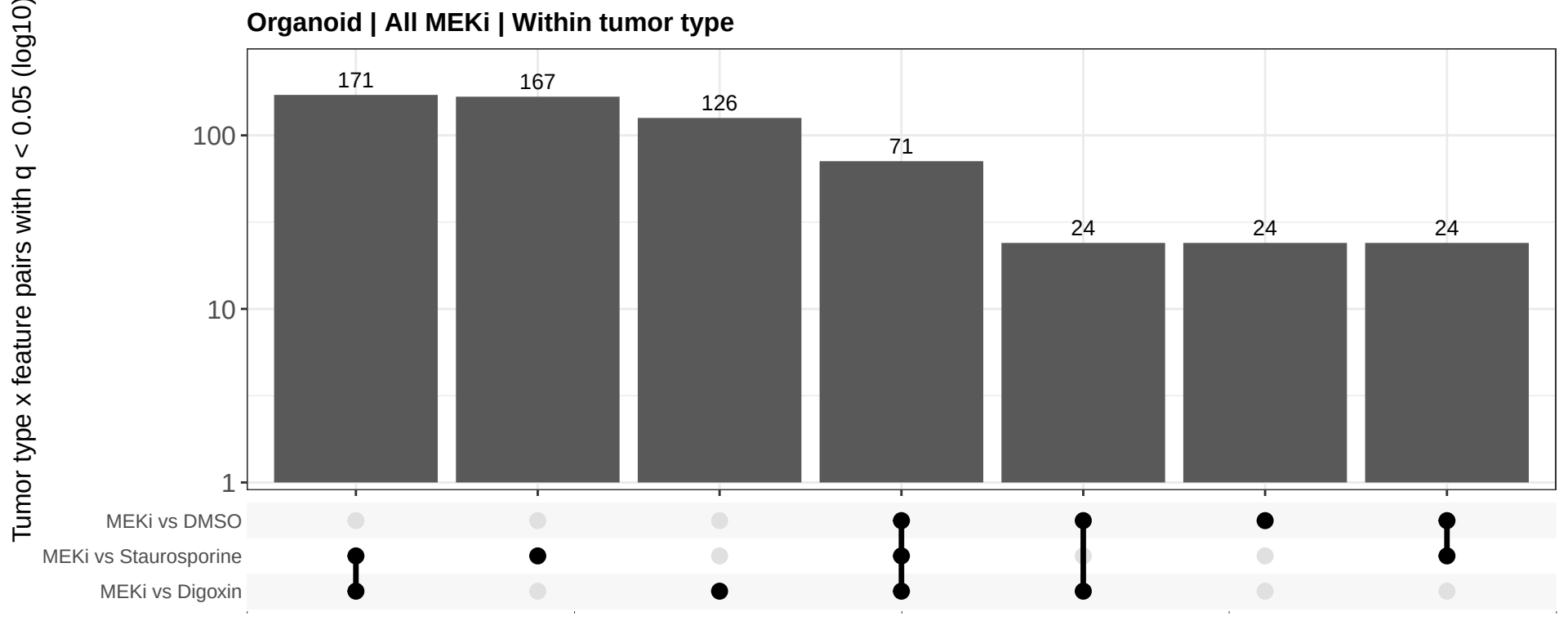

Organoid: features with q < 0.05
All MEKi
Binimetinib 1 uM
Binimetinib 10 uM
Mirdametinib 1 uM
Mirdametinib 10 uM
Trametinib 1 uM
Trametinib 10 uM
Selumetinib 1 uM
Selumetinib 10 uM
NF0014_T1 (cNF) 15 75 103 2 2 2 4 6 5 3 2 3 1 3 1 4 6 9 1 5 7 2 5 5 1 2 2
NF0018_T6 (cNF) 33 73 16 4 3 3 16 4 11 7 2 6 4 2 4 1 4 3 3 4 3 3 2 2 18 3 15
NF0021_T1 (cNF) 27 109 110 3 2 1 3 0 3 0 2 4 7 6 6 4 2 5 1 0 0 0 0 0 4 3 6
NF0035_T1 (cNF) 26 117 103 8 6 9 1 0 1 1 3 3 0 1 0 2 3 3 7 4 9 8 6 7 1 2 3
NF0037_T1 (cNF) 63 136 120 3 7 6 4 3 5 3 3 5 1 4 2 1 4 2 2 3 5 0 2 1 1 2 1
NF0014_T2 (pNF) 35 89 91 2 8 4 1 2 0 2 3 4 5 0 7 3 6 3 1 2 4 3 0 3 1 1 1
NF0016_T1 (pNF) 46 135 2 3 1 2 4 5 1 1 3 3 2 5 4 6 0 5
NF0055_T1 (pNF) 41 96 68 5 5 2 1 1 1 5 5 7 1 1 1 2 1 2 7 8 8 2 5 0 1 3 2
SARCO219_T2 (MPNST) 1 19 59 0 1 4 4 3 3 2 3 6 1 2 3 2 1 6 4 4 7 0 0 0 2 2 3
SARCO361_T1 (MPNST) 115 139 139 4 4 2 5 7 1 3 6 1 1 2 1 0 5 0 1 3 2 1 3 2 5 5 3
NF0030_T1 (Other) 6 112 132 6 13 14 4 5 2 3 2 4 2 5 10 3 6 7 2 3 5 3 4 7 17 1 23
NF0040_T1 (Other) 27 122 109 2 3 1 0 1 2 3 7 5 0 0 0 1 2 1 2 5 5 0 0 0 3 6 3
All patients 0 20 0 0 0 0 0 8 0 0 0 0 0 3 4 0 0 0 0 0 0 0 4 0 6 4 3
cNF 44 116 100 0 18 1 0 2 0 0 6 4 0 1 2 0 2 0 0 2 0 0 8 0 0 6 7
pNF 44 117 87 0 1 0 0 1 0 0 0 0 0 0 0 1 0 0 0 0 0 0 0 0 0 0 1
MPNST 34 86 84 0 0 0 0 1 0 0 0 1 0 0 0 0 0 1 0 0 0 0 0 0 0 0 0
Other 21 114 121 0 0 0 0 0 0 0 7 0 0 1 1 0 0 0 0 0 0 0 0 0 1 7 0
Tumor type means 0 0 0 0 0 0 1 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
Between tumor types 0 0 0 0 2 0 0 1 0 0 0 0 0 0 0 0 0 0 0 1 0 0 0 0 0 0 0
Within patient
Across patients
Within tumor type
Across tumor types (shared)
Across tumor types (differ)
MEKi vs DMSO
MEKi vs Staurosporine
MEKi vs Digoxin
Significant features
100
50
0


Single cell: features with q < 0.05
All MEKi
Binimetinib 1 uM
Binimetinib 10 uM
Mirdametinib 1 uM
Mirdametinib 10 uM
Trametinib 1 uM
Trametinib 10 uM
Selumetinib 1 uM
Selumetinib 10 uM
NF0014_T1 (cNF) 142 236 210 8 13 18 4 13 15 2 17 21 1 6 10 3 7 12 1 1 4 0 8 15 0 2 10
NF0018_T6 (cNF) 126 172 143 4 5 7 4 2 6 1 2 3 5 2 9 7 3 5 4 2 4 2 4 4 1 2 2
NF0021_T1 (cNF) 41 275 249 3 4 7 13 20 16 4 4 6 5 9 9 4 3 8 4 6 6 7 8 7 6 4 7
NF0035_T1 (cNF) 4 263 216 4 5 4 2 4 3 3 12 11 0 2 0 8 9 9 11 9 10 3 2 3 6 12 13
NF0037_T1 (cNF) 65 277 185 4 16 13 2 2 0 4 3 5 2 4 5 3 4 4 0 1 3 3 11 9 3 7 5
NF0014_T2 (pNF) 17 195 133 3 5 3 3 4 4 4 3 4 5 3 2 4 3 4 1 1 2 3 2 3 2 5 1
NF0016_T1 (pNF) 47 262 3 5 8 13 6 11 1 1 6 6 12 16 11 15 4 5
NF0055_T1 (pNF) 15 221 116 3 6 5 4 4 2 5 1 4 2 2 1 4 9 9 4 10 3 5 9 8 4 5 6
SARCO219_T2 (MPNST) 14 171 182 2 3 5 0 1 2 5 8 11 6 5 13 4 2 10 2 2 4 3 0 7 2 3 4
SARCO361_T1 (MPNST) 171 304 251 3 1 2 7 9 9 0 2 1 1 0 0 4 3 2 5 6 5 3 7 5 2 1 2
NF0030_T1 (Other) 2 200 259 5 4 7 4 11 9 2 6 7 1 4 3 6 7 6 1 6 12 6 3 10 13 5 26
NF0040_T1 (Other) 30 224 220 2 6 2 11 15 10 7 8 6 0 2 0 1 2 1 1 7 2 1 2 1 0 2 0
Within patient
All patients 0 83 24 0 35 0 0 0 0 0 51 5 0 19 2 0 18 0 0 60 10 0 48 0 0 80 11
Across patients
cNF 153 286 224 0 123 3 2 82 36 3 119 57 0 56 13 0 49 0 0 6 0 0 130 37 0 119 11
pNF 75 227 233 0 0 0 0 0 0 0 0 0 0 0 0 0 1 0 0 0 0 0 0 1 0 0 0
MPNST 38 278 225 0 0 0 0 0 2 0 0 0 0 0 1 0 0 0 0 0 0 0 0 0 1 0 0
Other 13 255 189 0 9 0 0 1 0 0 4 0 0 0 1 0 1 0 0 2 0 0 0 0 0 1 0
Within tumor type
Tumor type means 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
Across tumor types (shared)
Between tumor types 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
Across tumor types (differ)
MEKi vs DMSO
MEKi vs Staurosporine
MEKi vs Digoxin
Significant features
300
200
100
0


Organoid | All MEKi | Within patient
Patient x feature pairs with q < 0.05 (log10)
457
423
365
199
106
66
64
100
10
1
MEKi vs DMSO
MEKi vs Staurosporine
MEKi vs Digoxin


Organoid | All MEKi | Across patients
No significant features for: MEKi vs DMSO, MEKi vs Digoxin
Features with q < 0.05 (log10)
20
10
3
1
MEKi vs Staurosporine

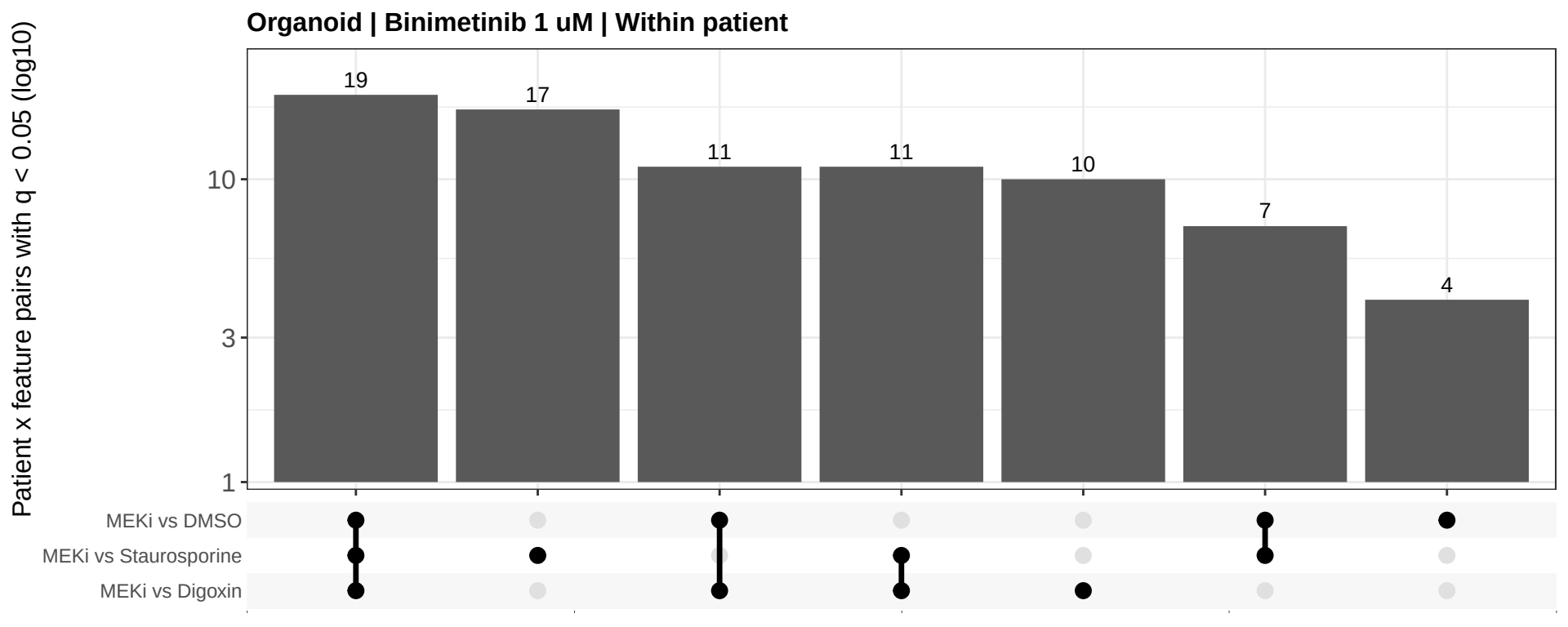

Organoid | All MEKi | Within tumor type
Tumor type x feature pairs with q < 0.05 (log10)
100
10
1
171
167
126
71
24
24
24
MEKi vs DMSO
MEKi vs Staurosporine
MEKi vs Digoxin

Organoid | All MEKi | Across tumor types (shared)

No features with q < 0.05

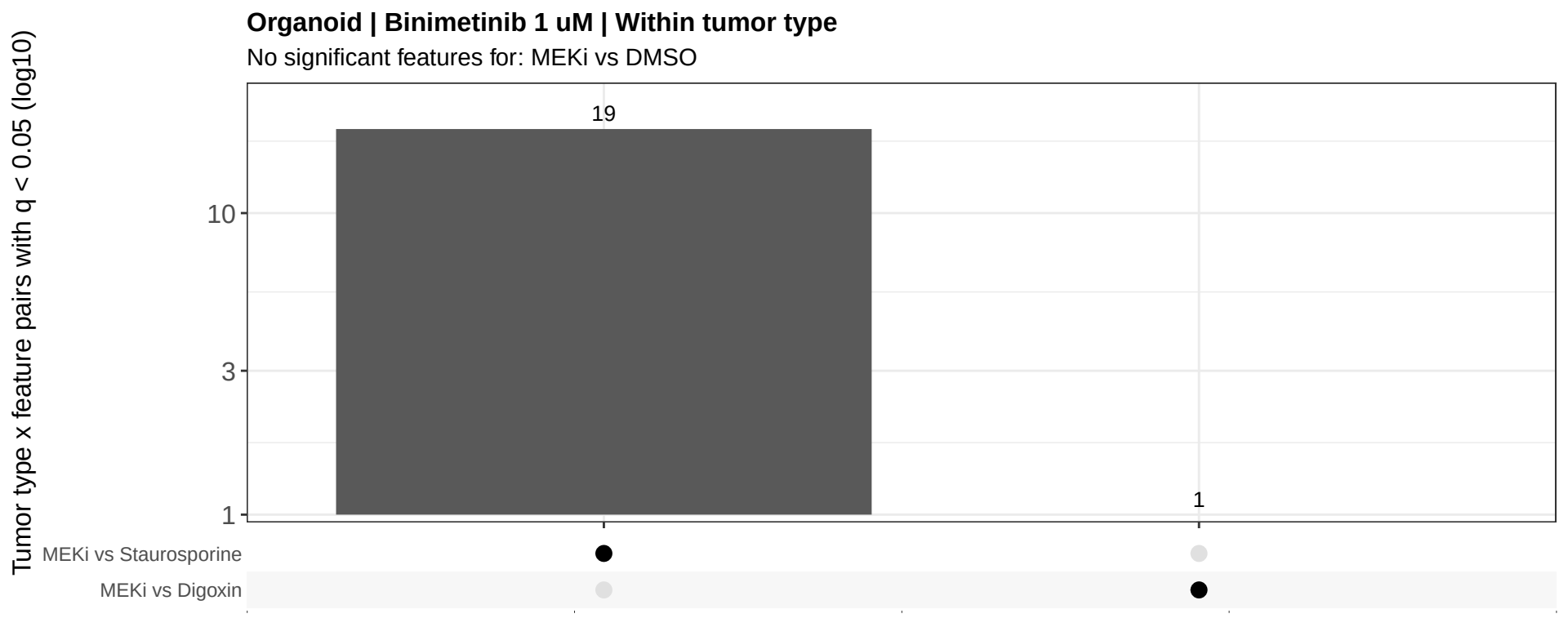
Organoid | All MEKi | Across tumor types (differ)

No features with q < 0.05


Organoid | Binimetinib 1 uM | Within patient
Patient x feature pairs with q < 0.05 (log10)
19
17
11
11
10
7
4
10
3
1
MEKi vs DMSO
MEKi vs Staurosporine
MEKi vs Digoxin

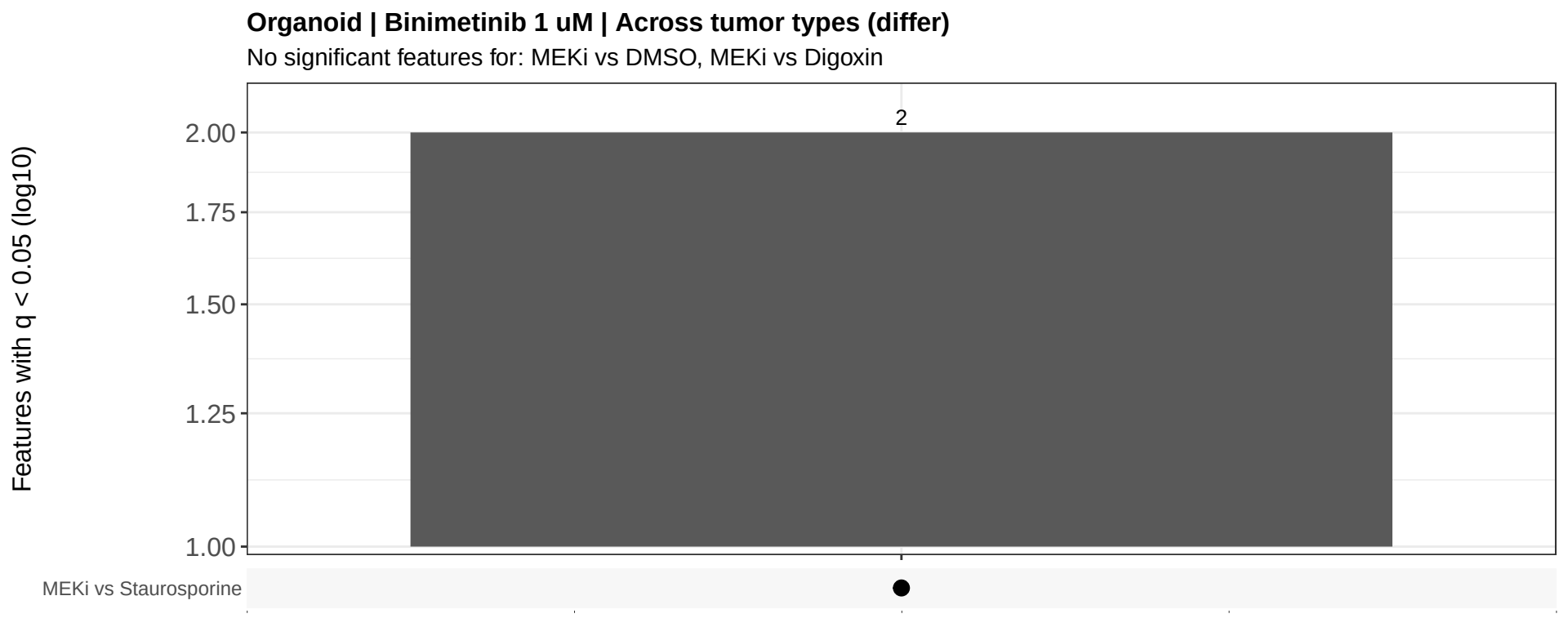
Organoid | Binimetinib 1 uM | Across patients

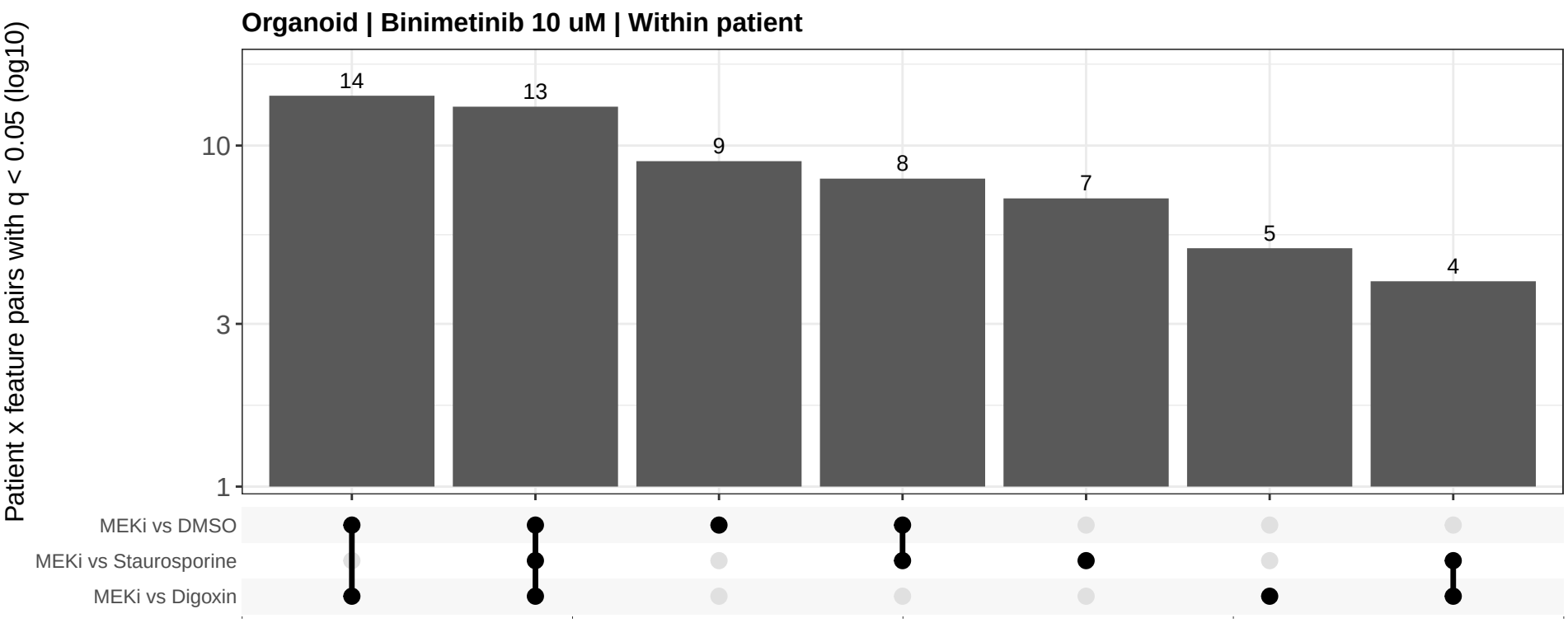
No features with q < 0.05

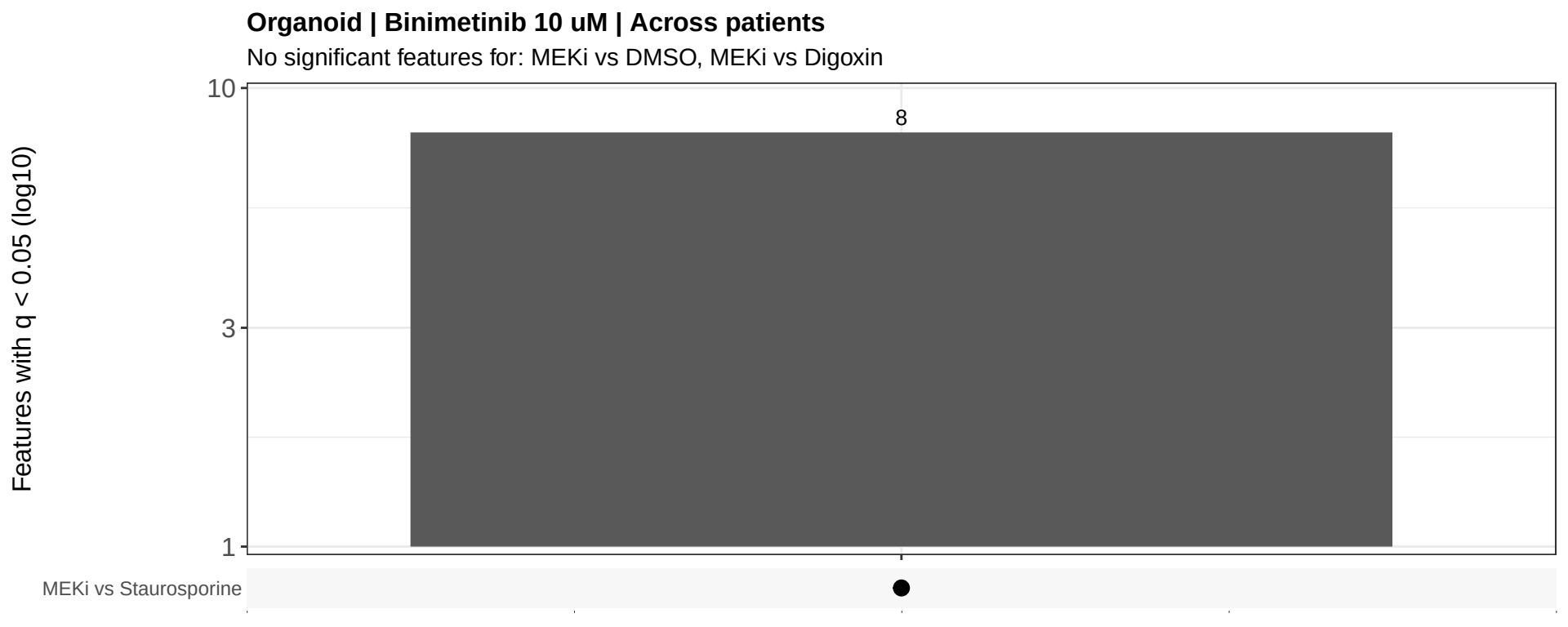

Organoid | Binimetinib 1 uM | Within tumor type
No significant features for: MEKi vs DMSO
Tumor type x feature pairs with q < 0.05 (log10)
19
1
10
3
1
MEKi vs Staurosporine
MEKi vs Digoxin

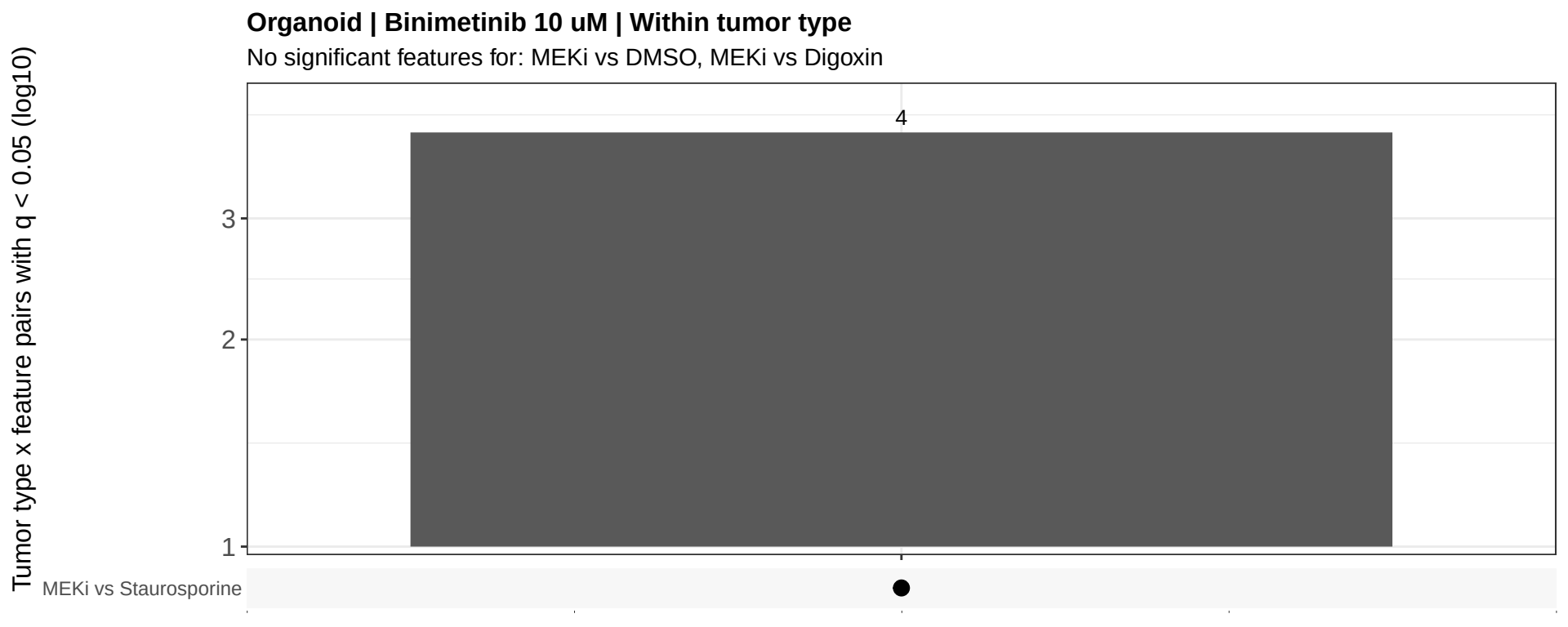
Organoid | Binimetinib 1 uM | Across tumor types (shared)

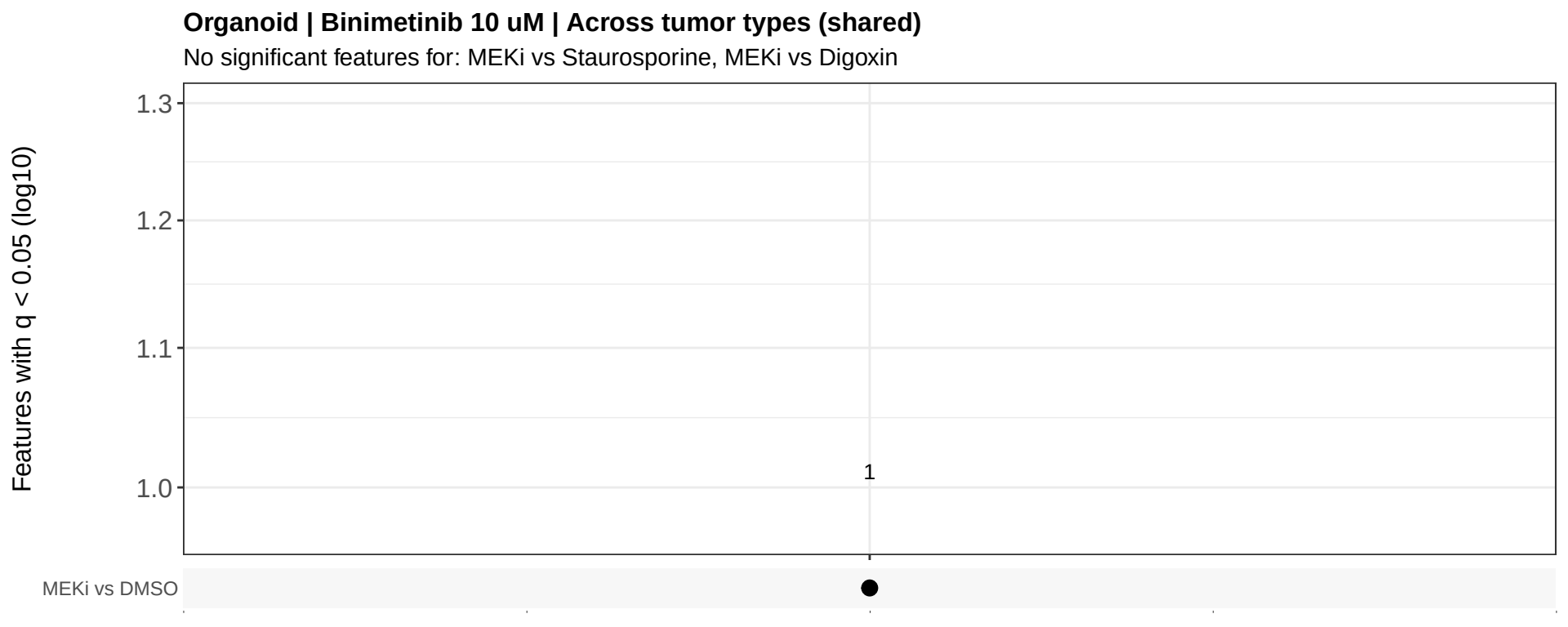
No features with q < 0.05

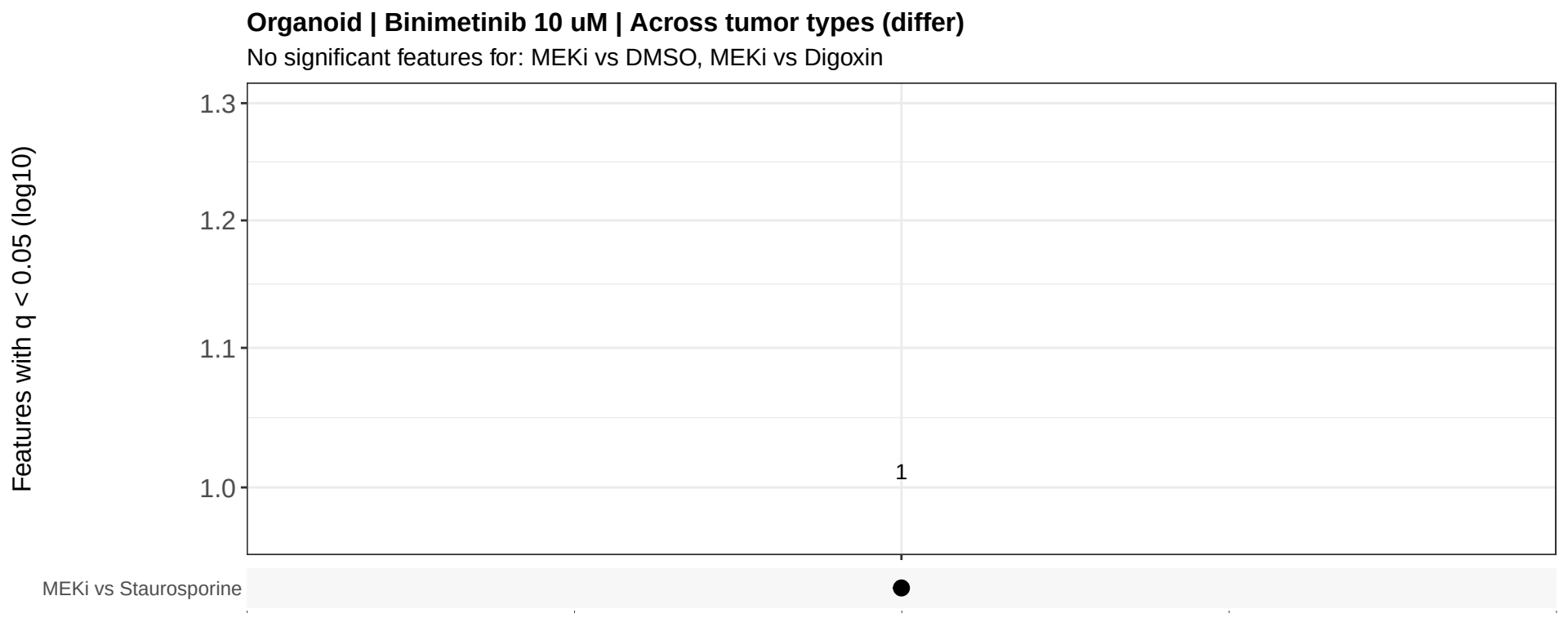

Organoid | Binimetinib 1 uM | Across tumor types (differ)
No significant features for: MEKi vs DMSO, MEKi vs Digoxin
Features with q < 0.05 (log10)
2.00
1.75
1.50
1.25
1.00
2
MEKi vs Staurosporine

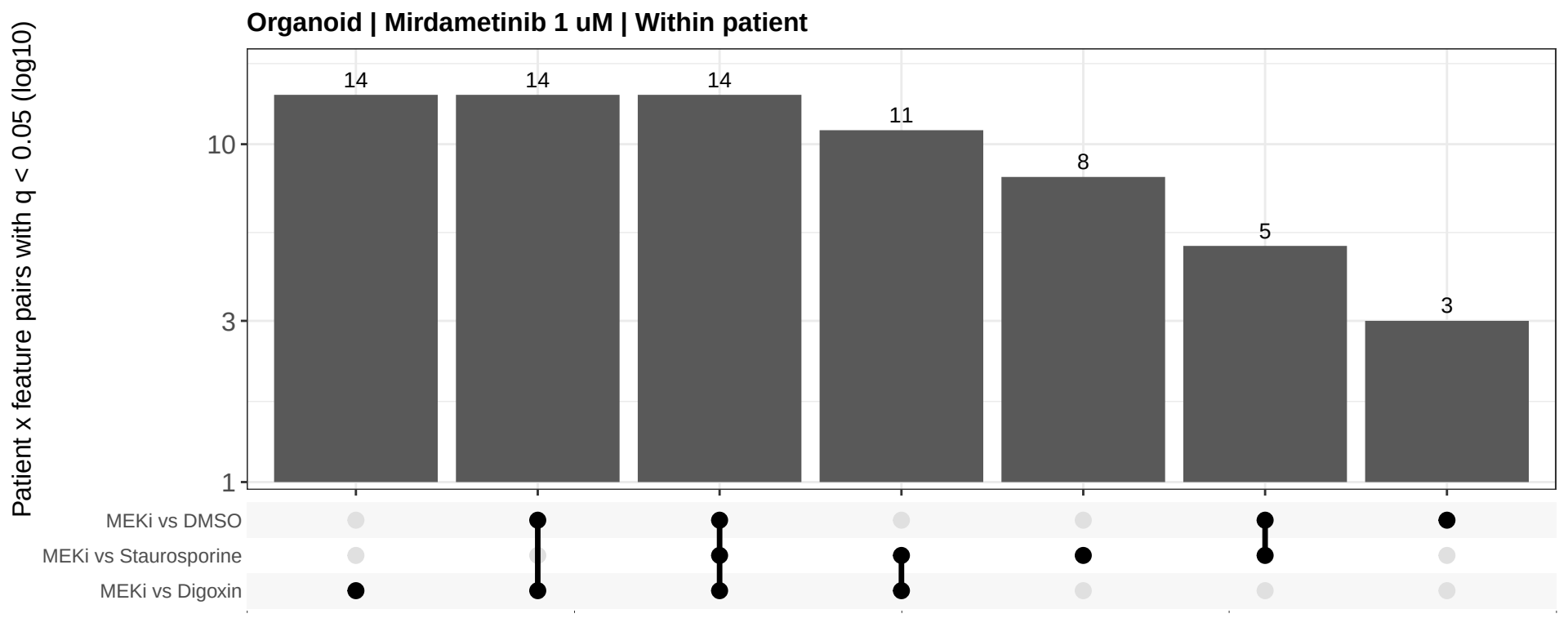

Organoid | Binimetinib 10 uM | Within patient
Patient x feature pairs with q < 0.05 (log10)
10
3
1
14
13
9
8
7
5
4
MEKi vs DMSO
MEKi vs Staurosporine
MEKi vs Digoxin


Organoid | Binimetinib 10 uM | Across patients
No significant features for: MEKi vs DMSO, MEKi vs Digoxin
Features with q < 0.05 (log10)
10
3
1
8
MEKi vs Staurosporine


Organoid | Binimetinib 10 uM | Within tumor type
No significant features for: MEKi vs DMSO, MEKi vs Digoxin
Tumor type x feature pairs with q < 0.05 (log10)
4
3
2
1
MEKi vs Staurosporine

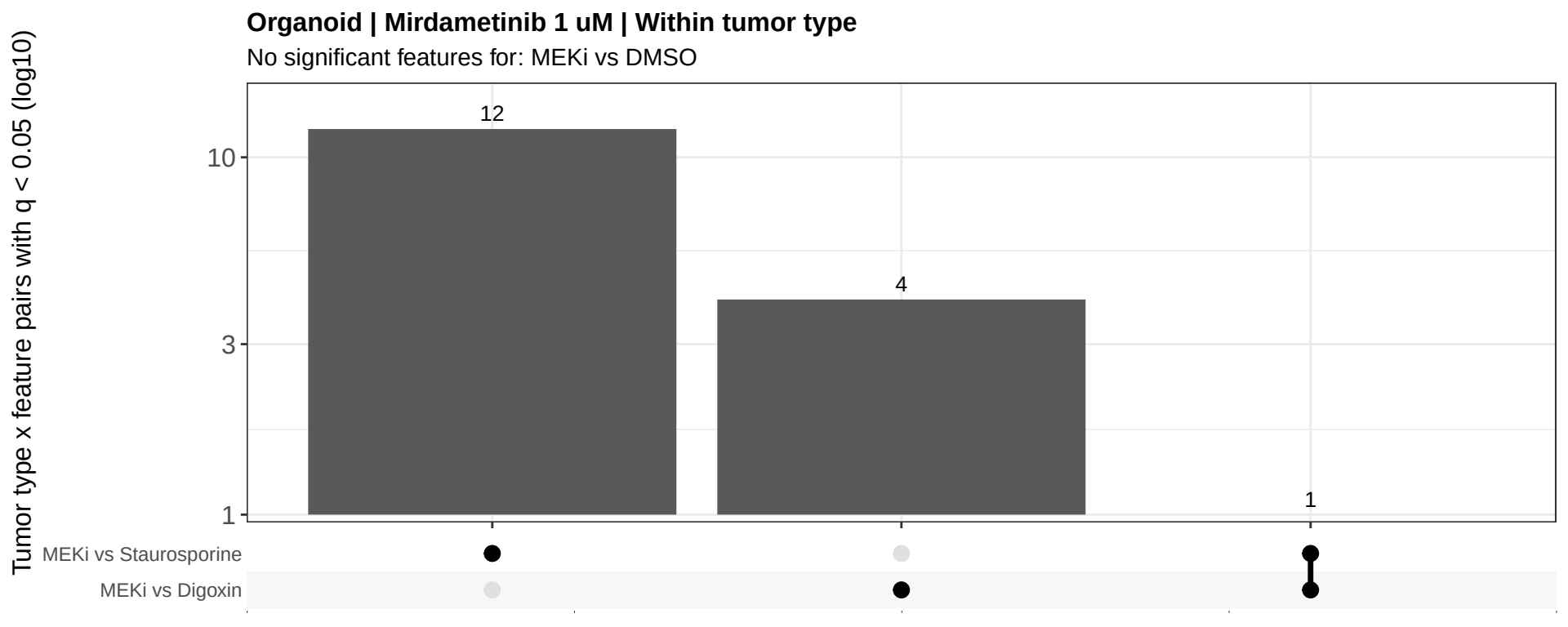

Organoid | Binimetinib 10 uM | Across tumor types (shared)
No significant features for: MEKi vs Staurosporine, MEKi vs Digoxin
Features with q < 0.05 (log10)
1.3
1.2
1.1
1.0
1
MEKi vs DMSO


Organoid | Binimetinib 10 uM | Across tumor types (differ)
No significant features for: MEKi vs DMSO, MEKi vs Digoxin
Features with q < 0.05 (log10)
1.3
1.2
1.1
1.0
1
MEKi vs Staurosporine


Organoid | Mirdametinib 1 uM | Within patient
Patient x feature pairs with q < 0.05 (log10)
14
14
14
11
8
5
3
10
3
1
MEKi vs DMSO
MEKi vs Staurosporine
MEKi vs Digoxin

Organoid | Mirdametinib 1 uM | Across patients

No features with q < 0.05

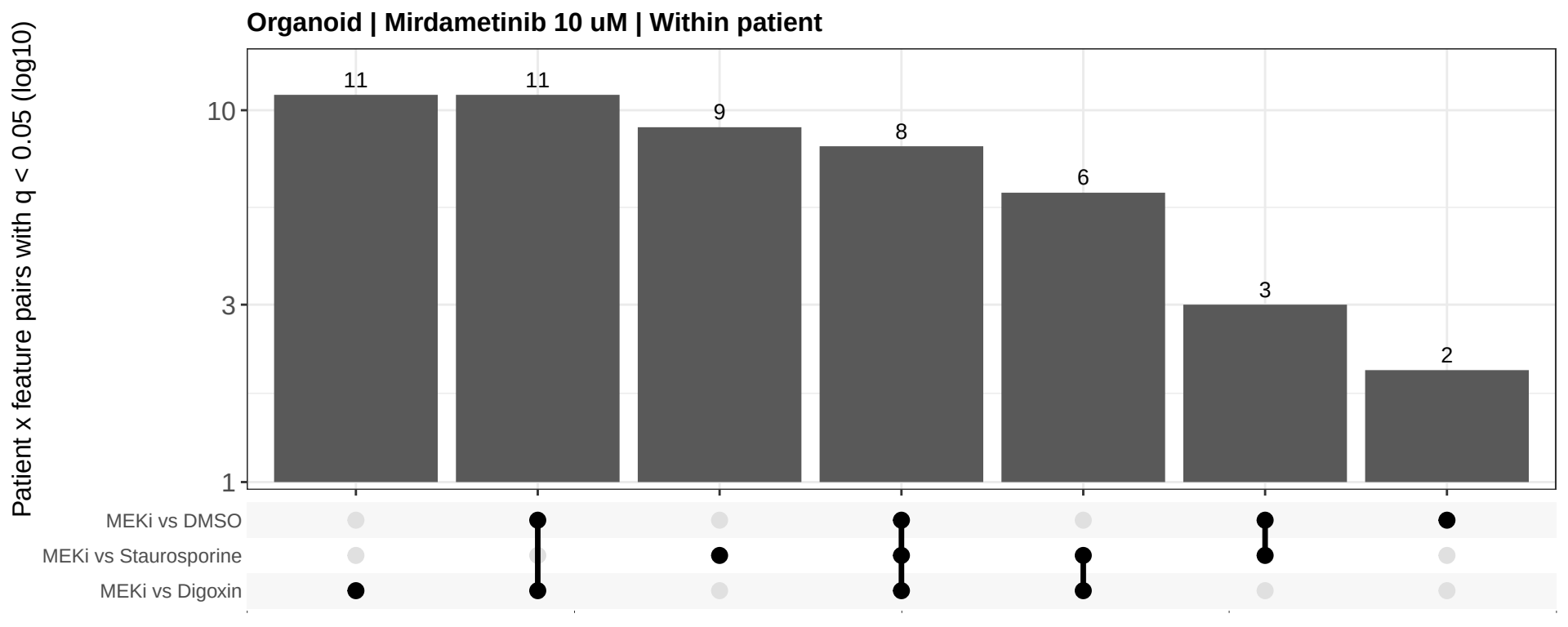

Organoid | Mirdametinib 1 uM | Within tumor type
No significant features for: MEKi vs DMSO
Tumor type x feature pairs with q < 0.05 (log10)
12
4
1
10
3
1
MEKi vs Staurosporine
MEKi vs Digoxin

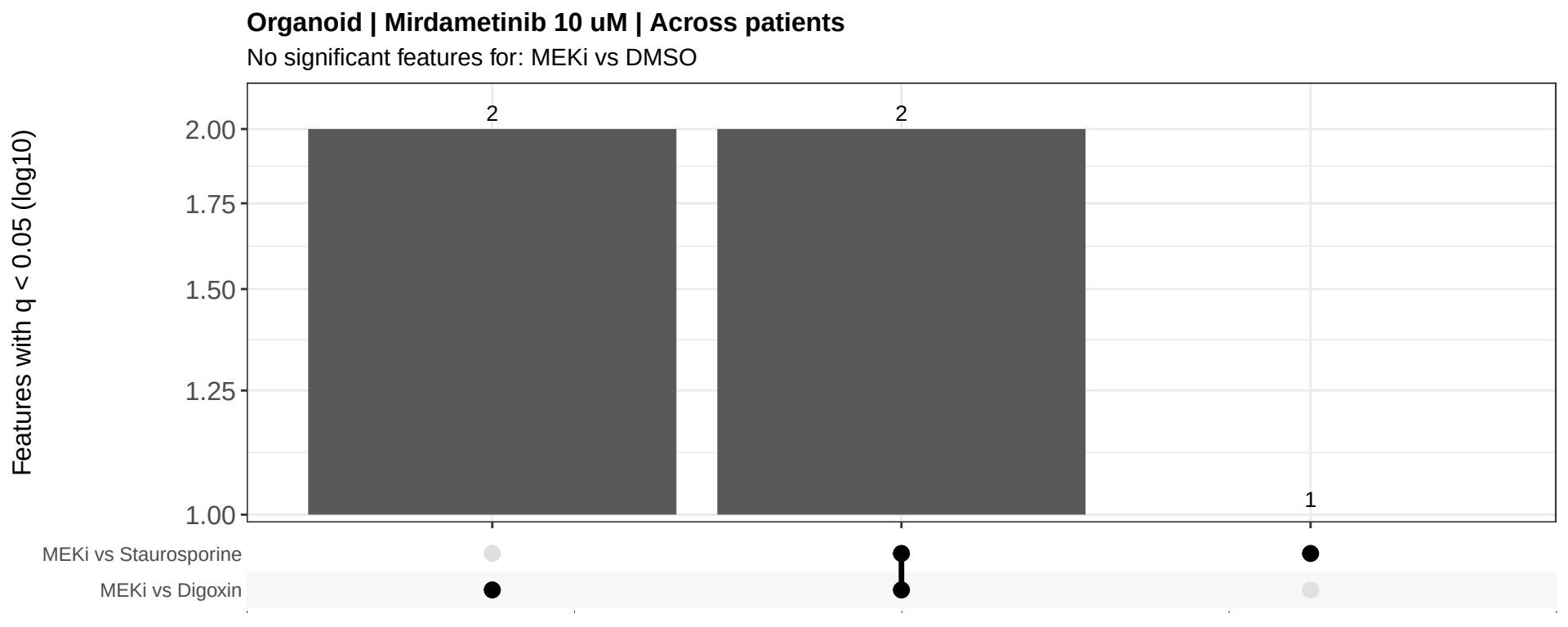
Organoid | Mirdametinib 1 uM | Across tumor types (shared)

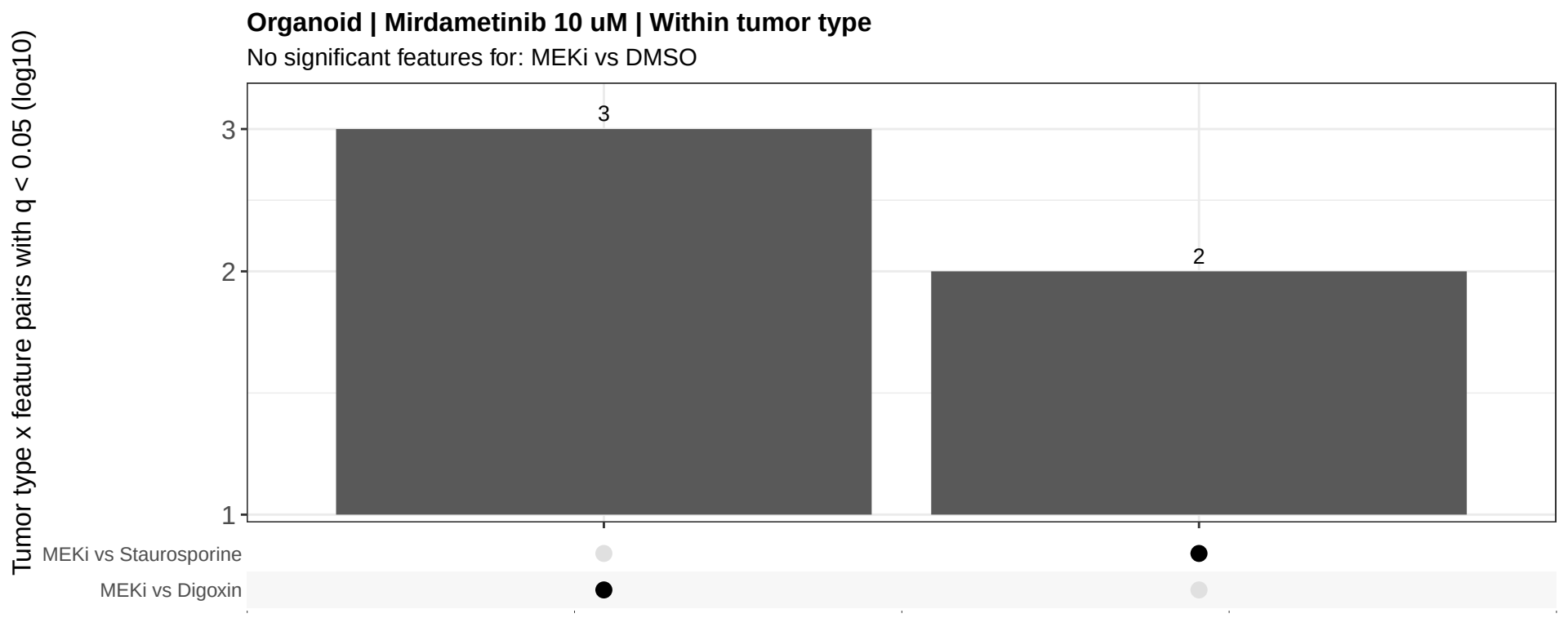
No features with q < 0.05

Organoid | Mirdametinib 1 uM | Across tumor types (differ)

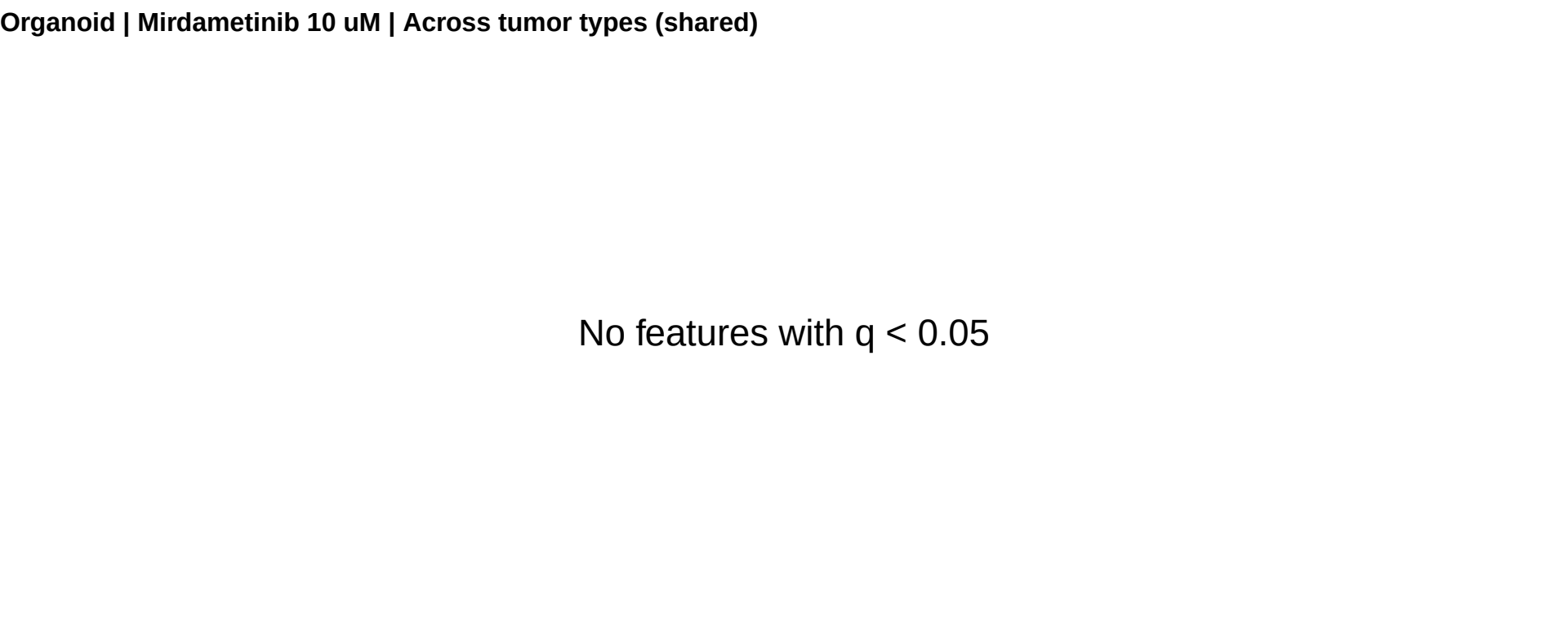
No features with q < 0.05


Organoid | Mirdametinib 10 uM | Within patient
Patient x feature pairs with q < 0.05 (log10)
10
3
1
11
11
9
8
6
3
2
MEKi vs DMSO
MEKi vs Staurosporine
MEKi vs Digoxin


Organoid | Mirdametinib 10 uM | Across patients
No significant features for: MEKi vs DMSO
Features with q < 0.05 (log10)
2.00
1.75
1.50
1.25
1.00
2
2
1
MEKi vs Staurosporine
MEKi vs Digoxin

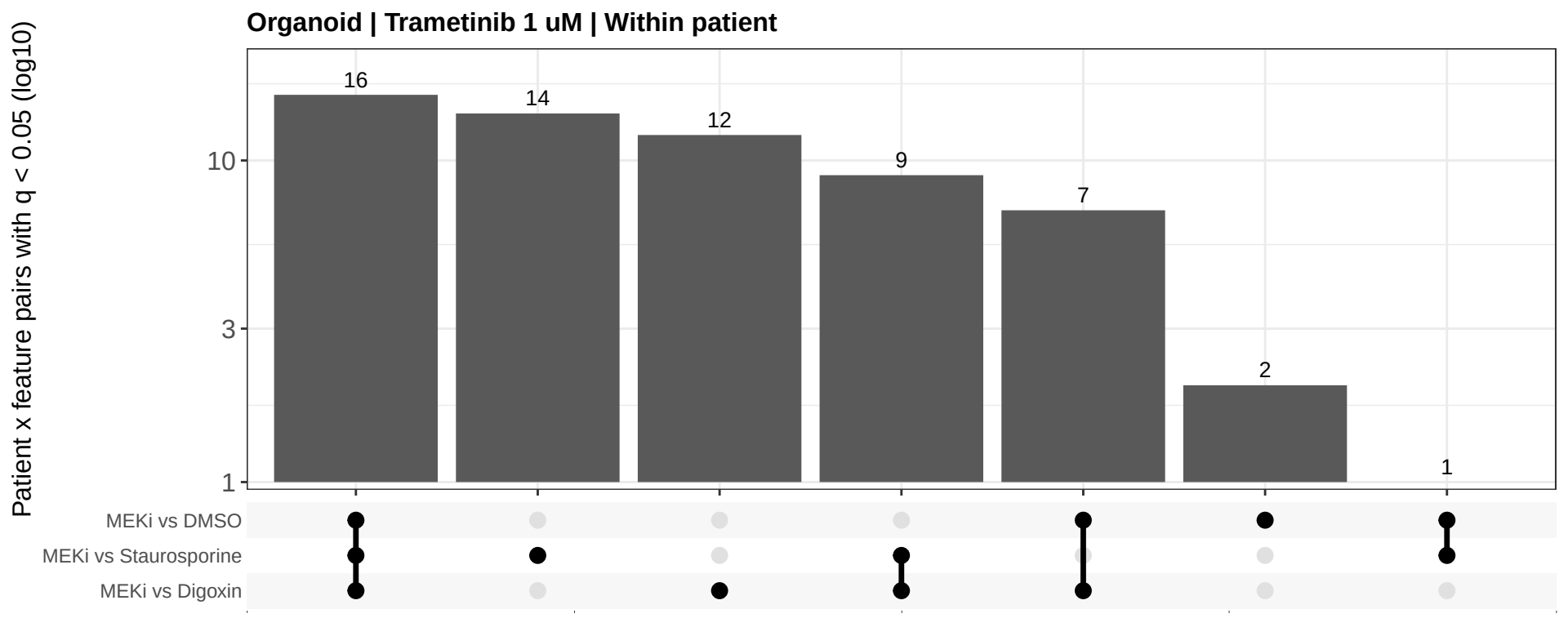

Organoid | Mirdametinib 10 uM | Within tumor type
No significant features for: MEKi vs DMSO
Tumor type x feature pairs with q < 0.05 (log10)
3
2
1
3
2
MEKi vs Staurosporine
MEKi vs Digoxin

Organoid | Mirdametinib 10 uM | Across tumor types (shared)



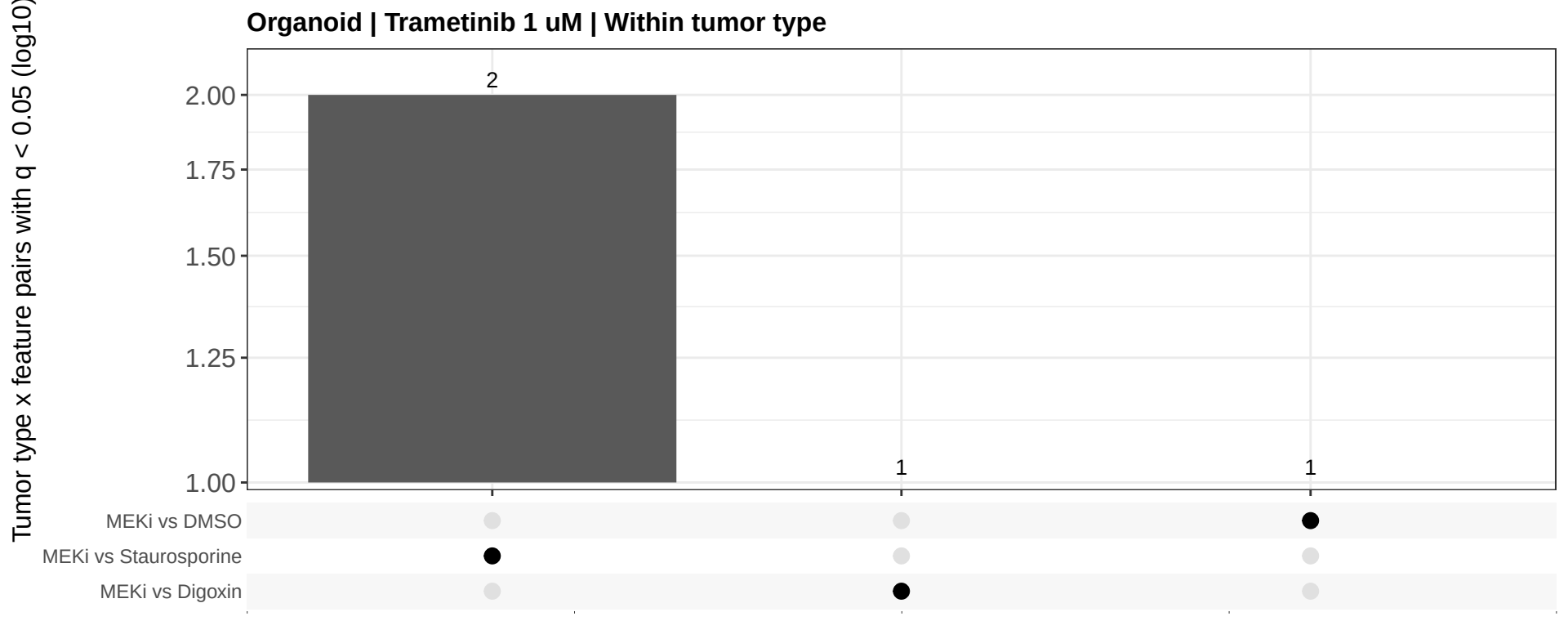
Organoid | Mirdametinib 10 uM | Across tumor types (differ)

No features with q < 0.05


Organoid | Trametinib 1 uM | Within patient
Patient x feature pairs with q < 0.05 (log10)
16
14
12
9
7
2
1
10
3
1
MEKi vs DMSO
MEKi vs Staurosporine
MEKi vs Digoxin

Organoid | Trametinib 1 uM | Across patients

No features with q < 0.05

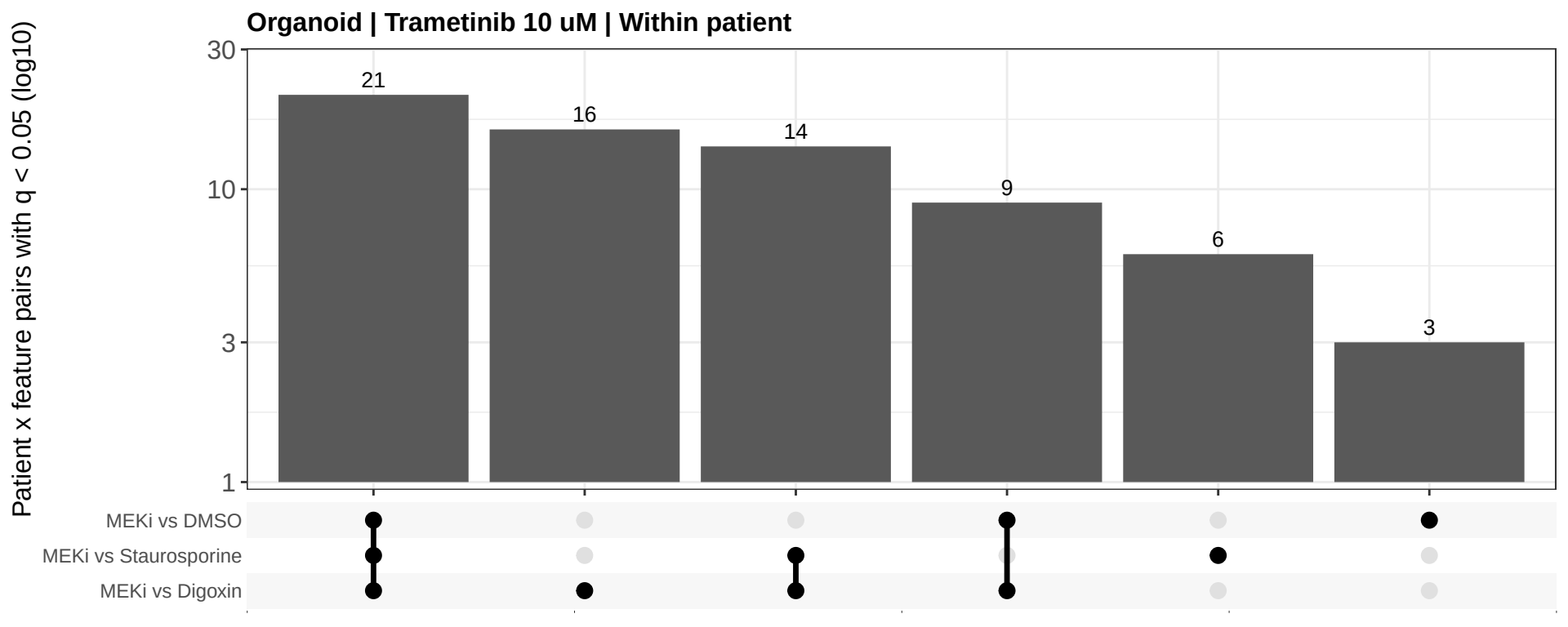

Organoid | Trametinib 1 uM | Within tumor type
Tumor type x feature pairs with q < 0.05 (log10)
2.00
1.75
1.50
1.25
1.00
2
1
1
MEKi vs DMSO
MEKi vs Staurosporine
MEKi vs Digoxin

Organoid | Trametinib 1 uM | Across tumor types (shared)

No features with q < 0.05

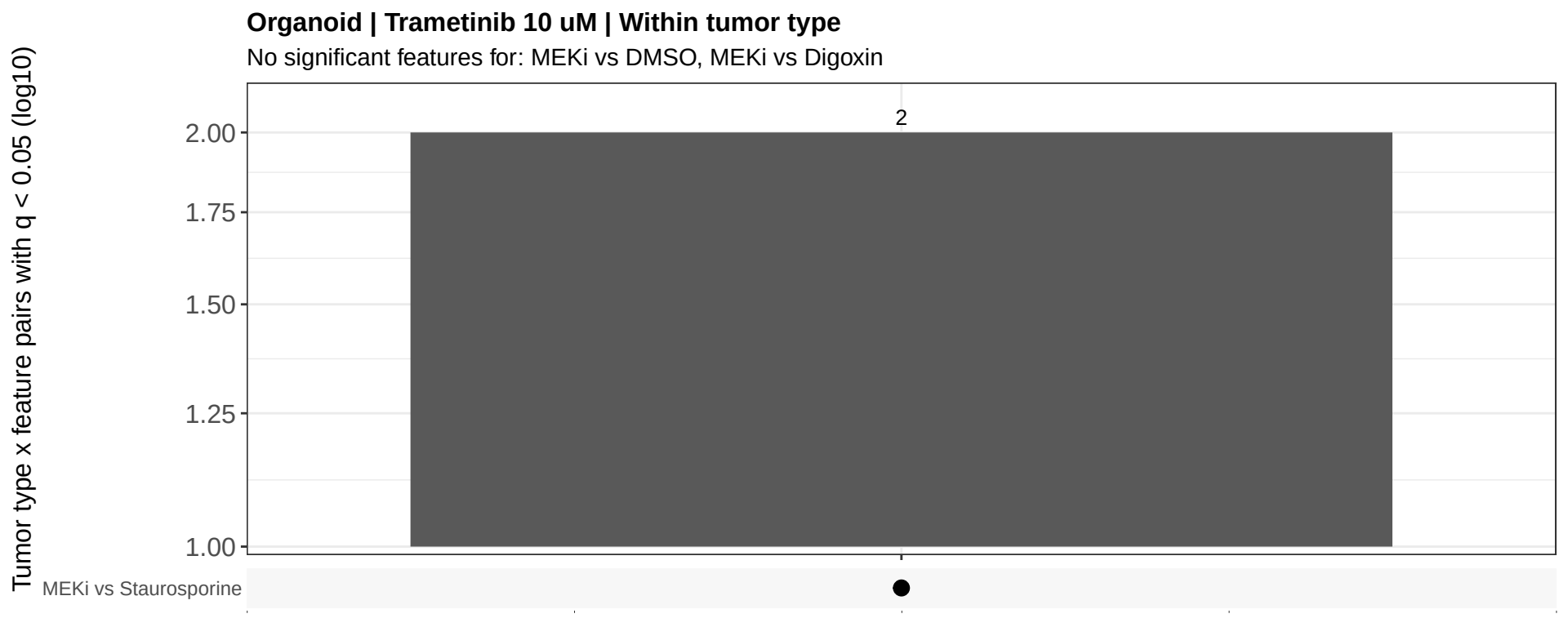
Organoid | Trametinib 1 uM | Across tumor types (differ)

No features with q < 0.05


Organoid | Trametinib 10 uM | Within patient
Patient x feature pairs with q < 0.05 (log10)
30
10
3
1
21
16
14
9
6
3
MEKi vs DMSO
MEKi vs Staurosporine
MEKi vs Digoxin

Organoid | Trametinib 10 uM | Across patients

No features with q < 0.05

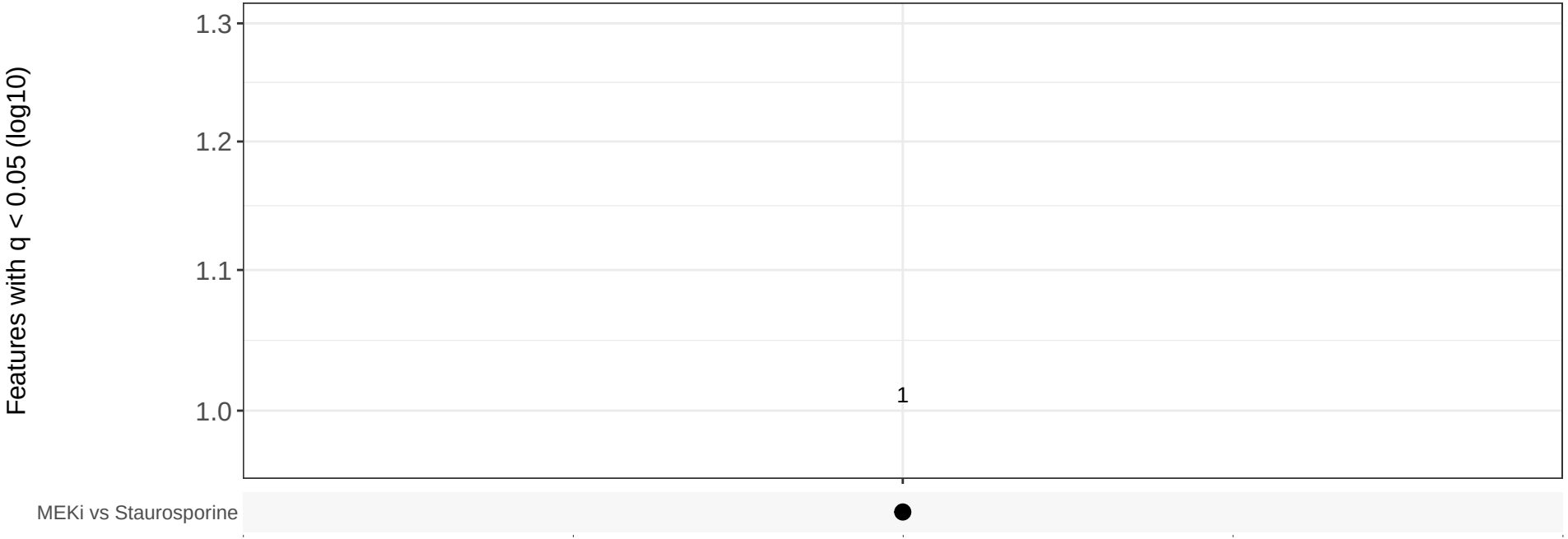

Organoid | Trametinib 10 uM | Within tumor type
No significant features for: MEKi vs DMSO, MEKi vs Digoxin
Tumor type x feature pairs with q < 0.05 (log10)
2.00
1.75
1.50
1.25
1.00
2
MEKi vs Staurosporine

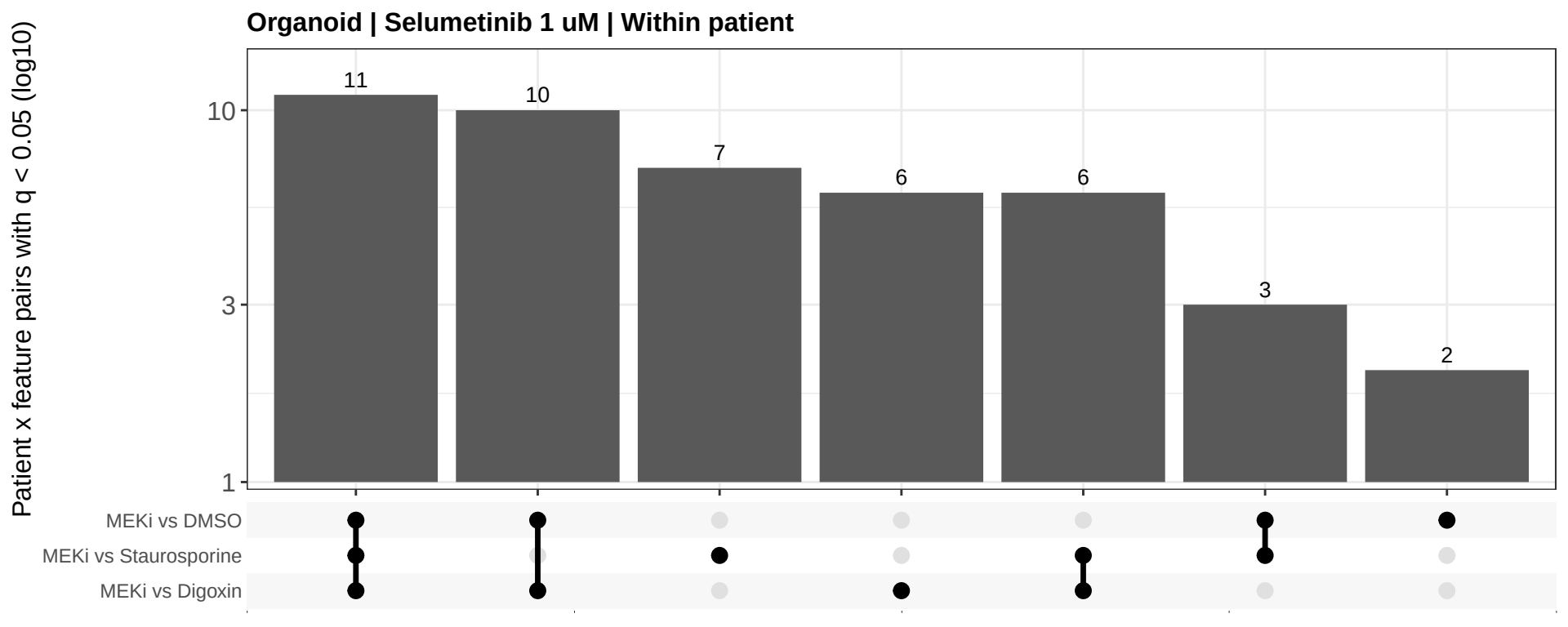
Organoid | Trametinib 10 uM | Across tumor types (shared)

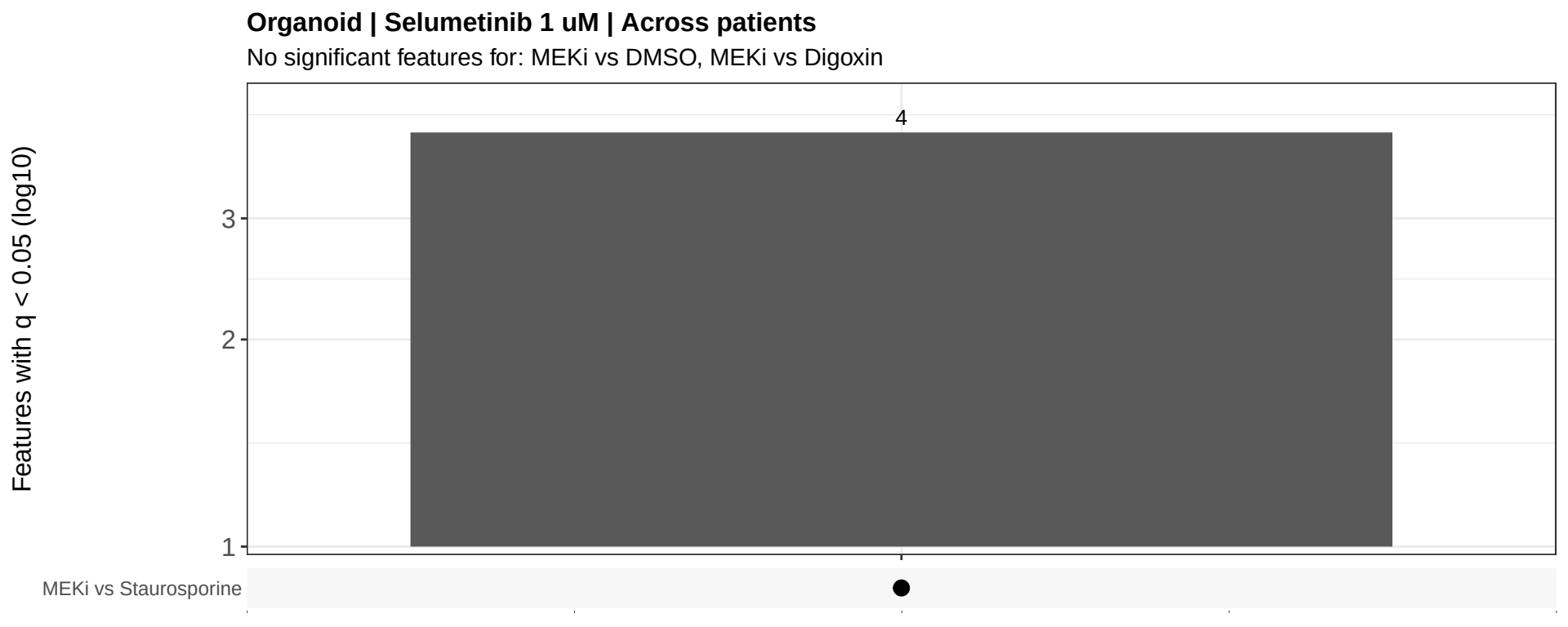
No features with q < 0.05

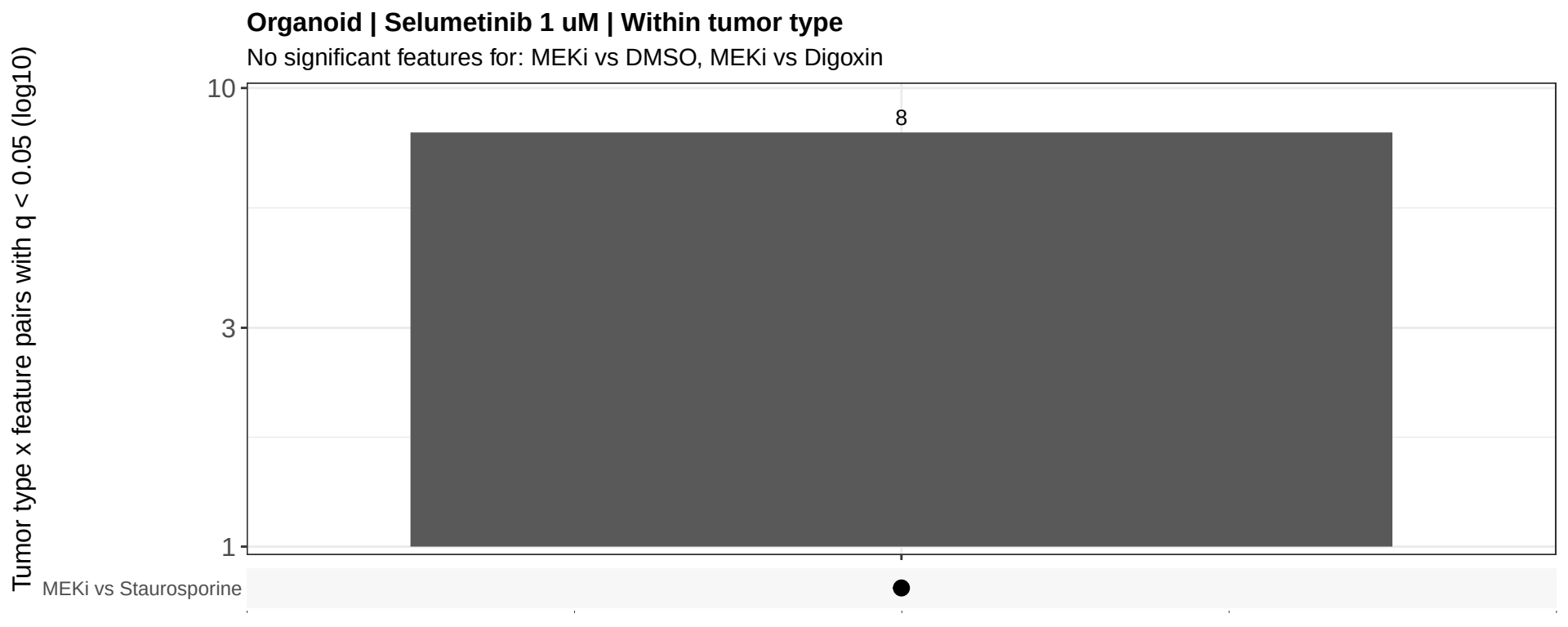
Organoid | Trametinib 10 uM | Across tumor types (differ)

No significant features for: MEKi vs DMSO, MEKi vs Digoxin




Organoid | Selumetinib 1 uM | Within patient
Patient x feature pairs with q < 0.05 (log10)
11
10
7
6
6
3
2
10
3
1
MEKi vs DMSO
MEKi vs Staurosporine
MEKi vs Digoxin


Organoid | Selumetinib 1 uM | Across patients
No significant features for: MEKi vs DMSO, MEKi vs Digoxin
Features with q < 0.05 (log10)
4
3
2
1
MEKi vs Staurosporine

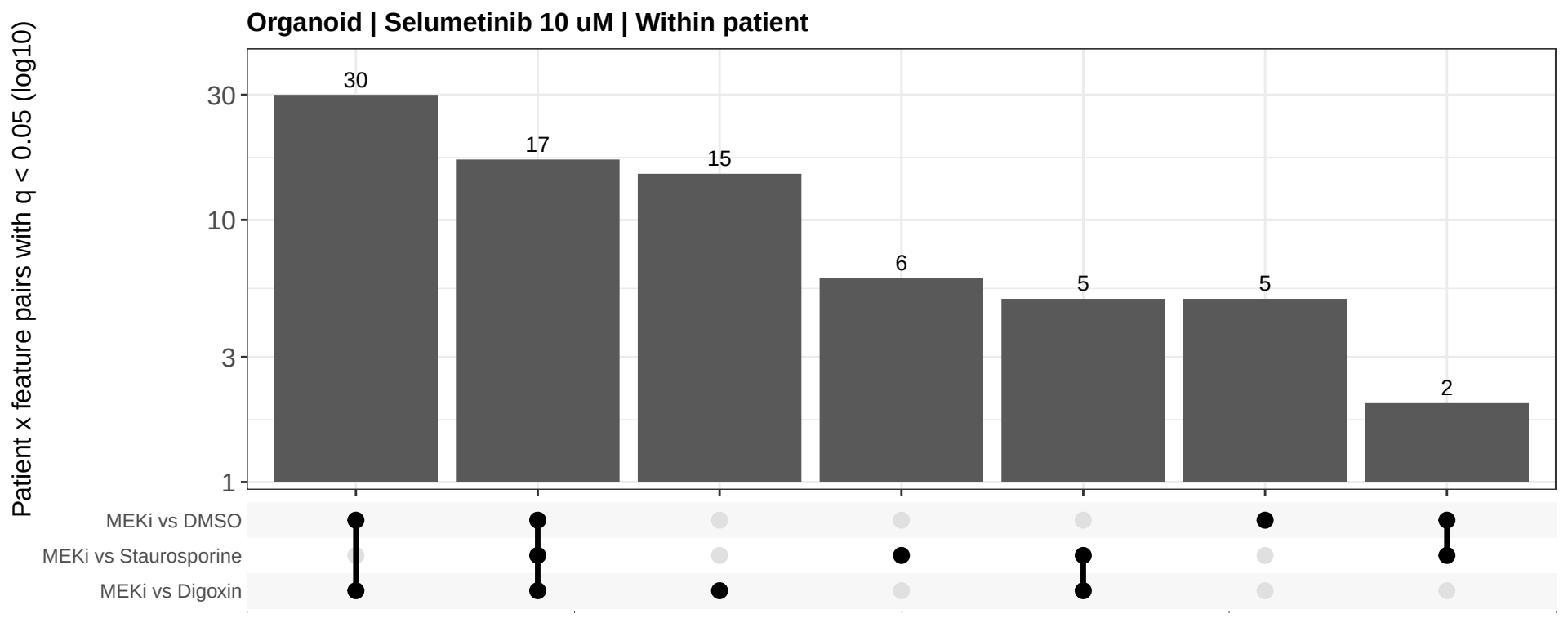

Organoid | Selumetinib 1 uM | Within tumor type
No significant features for: MEKi vs DMSO, MEKi vs Digoxin
Tumor type x feature pairs with q < 0.05 (log10)
10
3
1
8
MEKi vs Staurosporine

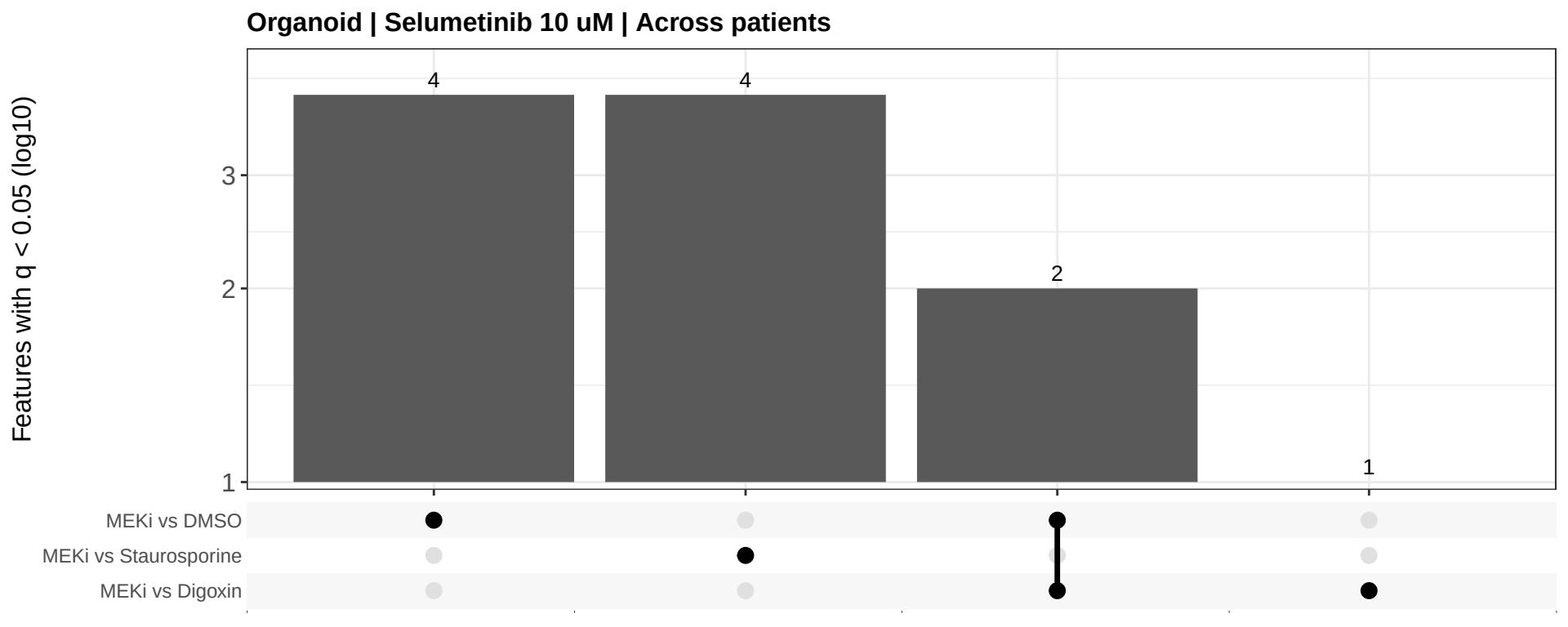
Organoid | Selumetinib 1 uM | Across tumor types (shared)

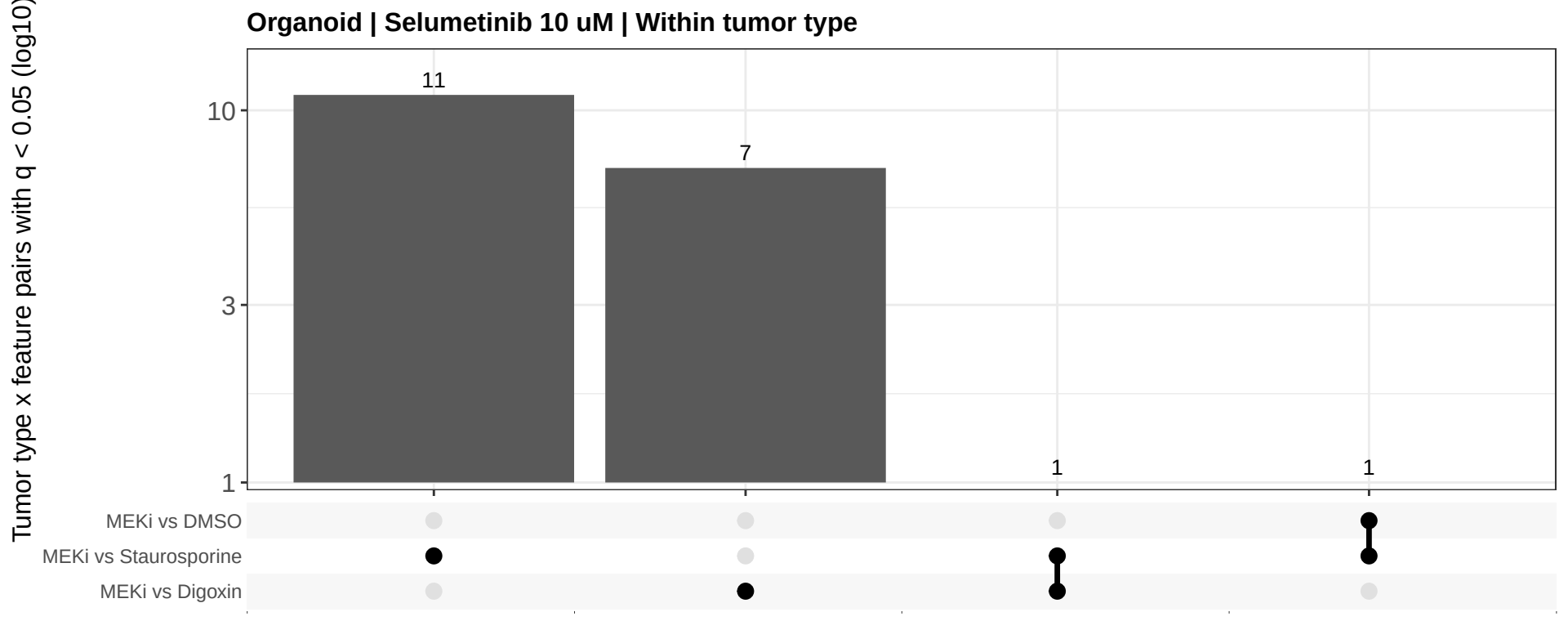
No features with q < 0.05

Organoid | Selumetinib 1 uM | Across tumor types (differ)

No features with q < 0.05


Organoid | Selumetinib 10 uM | Within patient
Patient x feature pairs with q < 0.05 (log10)
30
10
3
1
30
17
15
6
5
5
2
MEKi vs DMSO
MEKi vs Staurosporine
MEKi vs Digoxin


Organoid | Selumetinib 10 uM | Across patients
Features with q < 0.05 (log10)
4
4
2
1
3
2
1
MEKi vs DMSO
MEKi vs Staurosporine
MEKi vs Digoxin

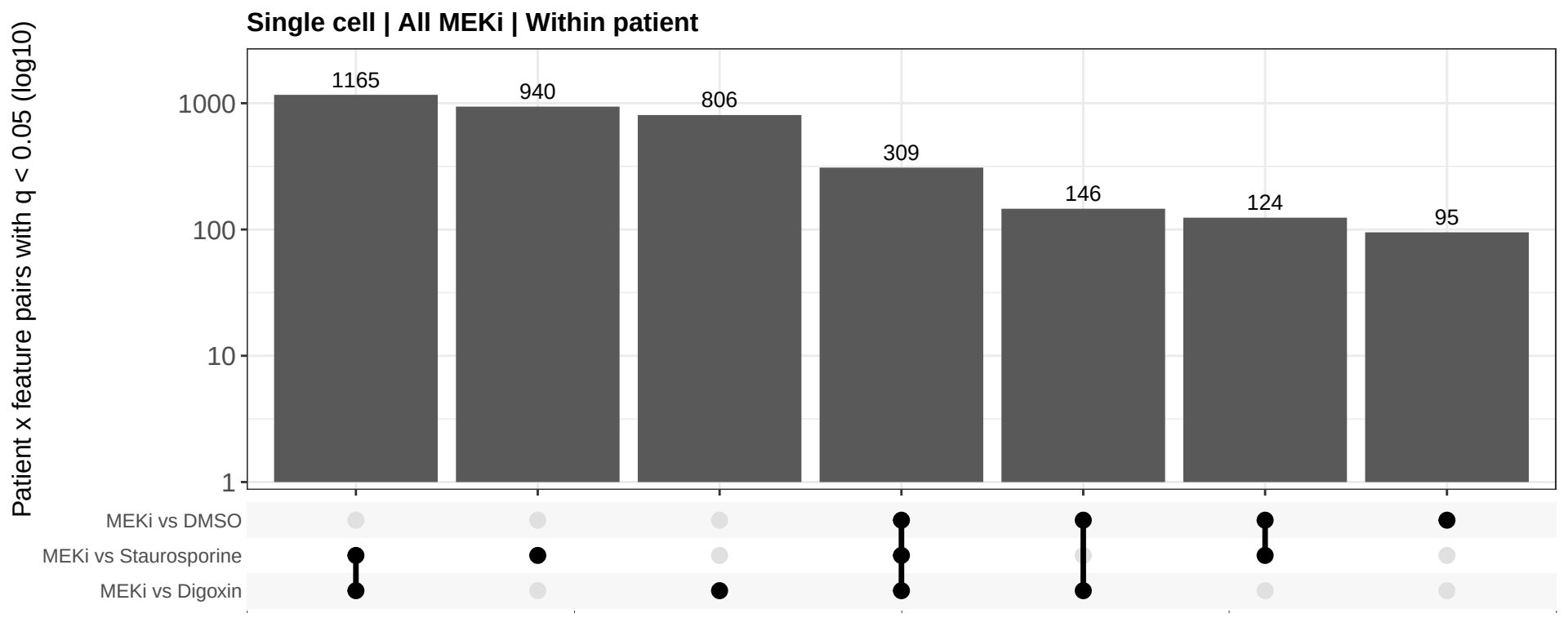

Organoid | Selumetinib 10 uM | Within tumor type
Tumor type x feature pairs with q < 0.05 (log10)
10
3
1
11
7
1
1
MEKi vs DMSO
MEKi vs Staurosporine
MEKi vs Digoxin

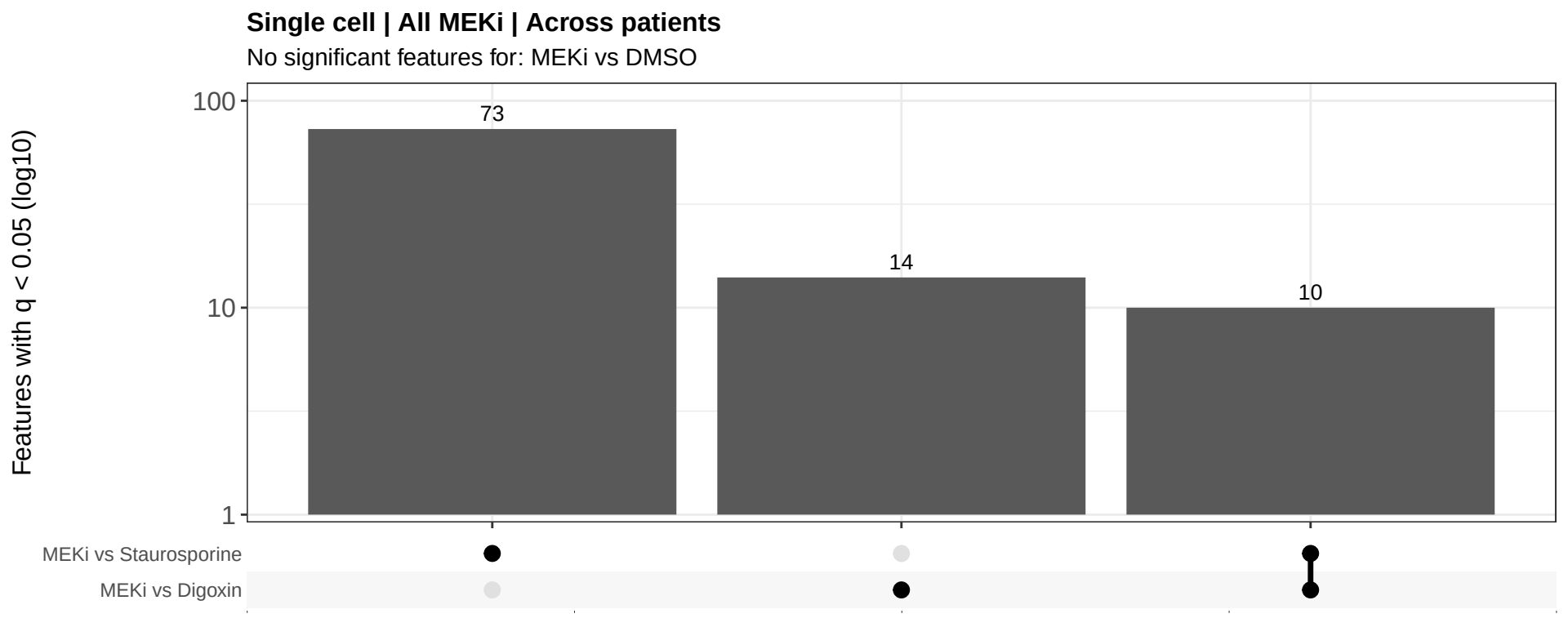
Organoid | Selumetinib 10 uM | Across tumor types (shared)

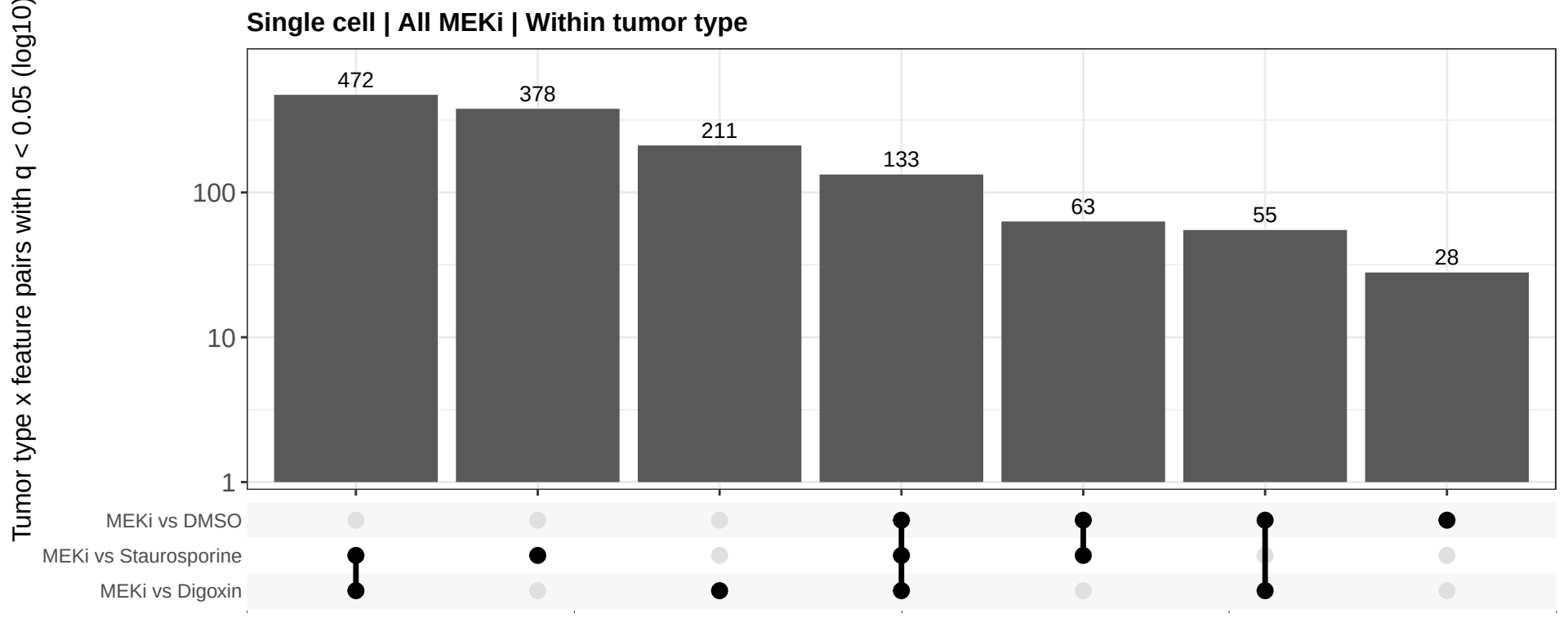
No features with q < 0.05

Organoid | Selumetinib 10 uM | Across tumor types (differ)

No features with q < 0.05


Single cell | All MEKi | Within patient
Patient x feature pairs with q < 0.05 (log10)
1000
100
10
1
1165
940
806
309
146
124
95
MEKi vs DMSO
MEKi vs Staurosporine
MEKi vs Digoxin


Single cell | All MEKi | Across patients
No significant features for: MEKi vs DMSO
Features with q < 0.05 (log10)
100
10
1
73
14
10
MEKi vs Staurosporine
MEKi vs Digoxin

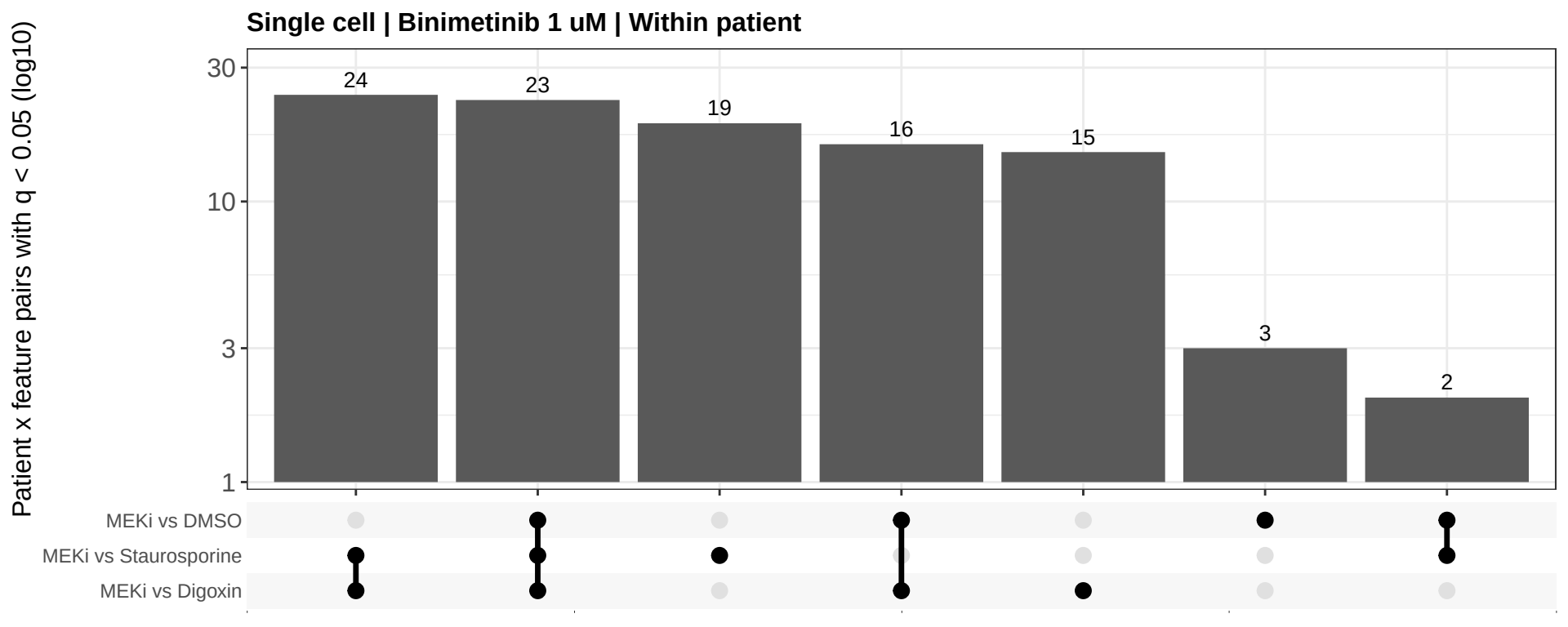

Single cell | All MEKi | Within tumor type
Tumor type x feature pairs with q < 0.05 (log10)
472
378
211
133
63
55
28
100
10
1
MEKi vs DMSO
MEKi vs Staurosporine
MEKi vs Digoxin

Single cell | All MEKi | Across tumor types (shared)

No features with q < 0.05

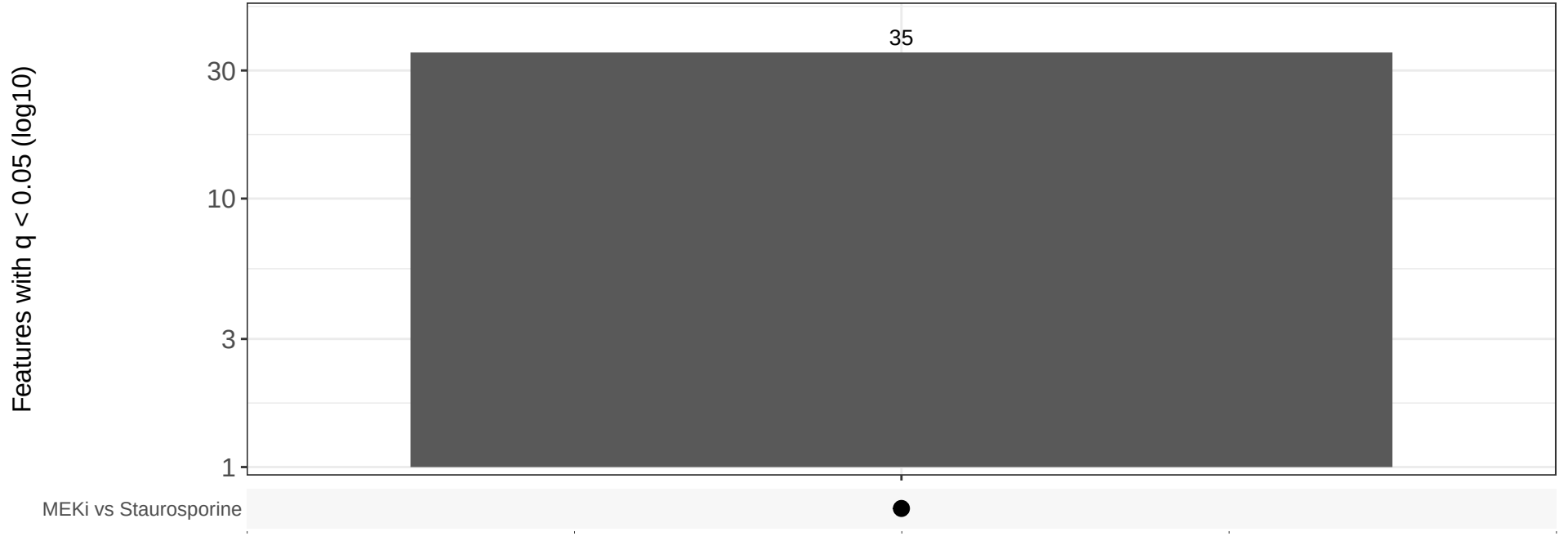
Single cell | All MEKi | Across tumor types (differ)

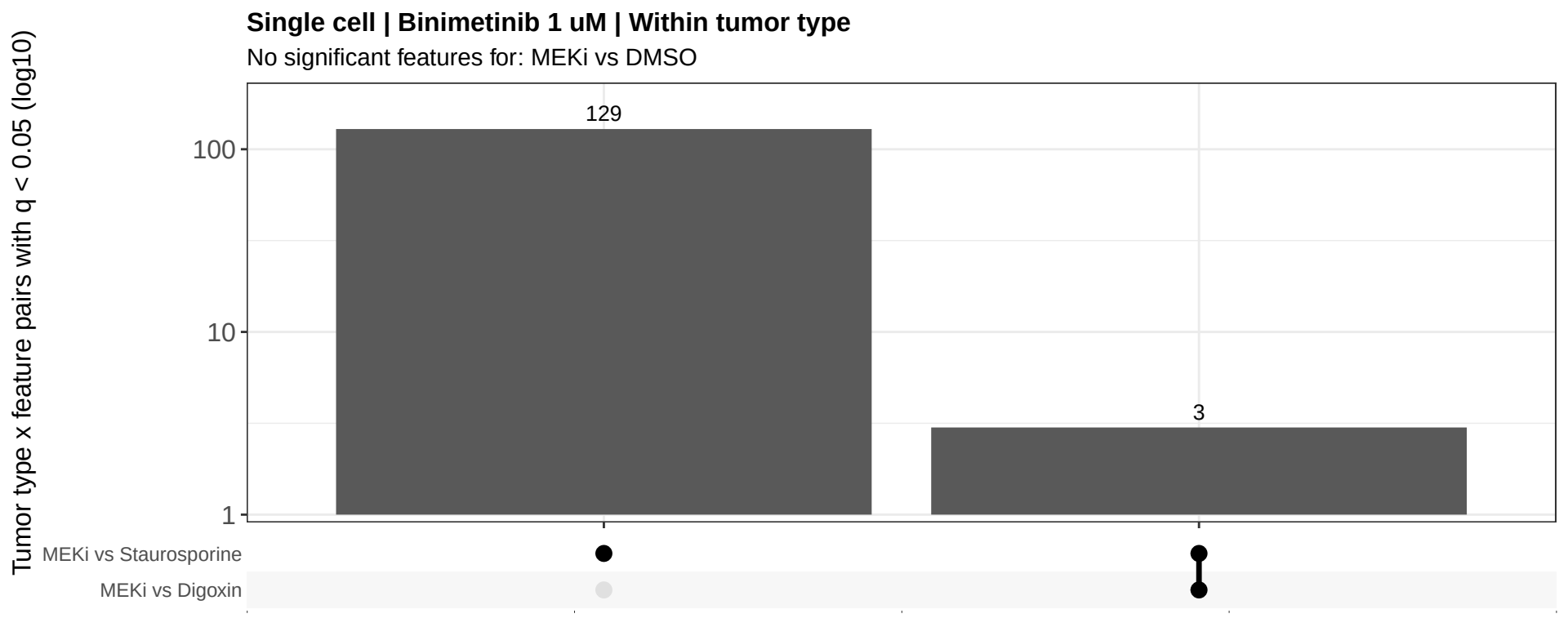
No features with q < 0.05


Single cell | Binimetinib 1 uM | Within patient
Patient x feature pairs with q < 0.05 (log10)
30
10
3
1
24
23
19
16
15
3
2
MEKi vs DMSO
MEKi vs Staurosporine
MEKi vs Digoxin

Single cell | Binimetinib 1 uM | Across patients

No significant features for: MEKi vs DMSO, MEKi vs Digoxin



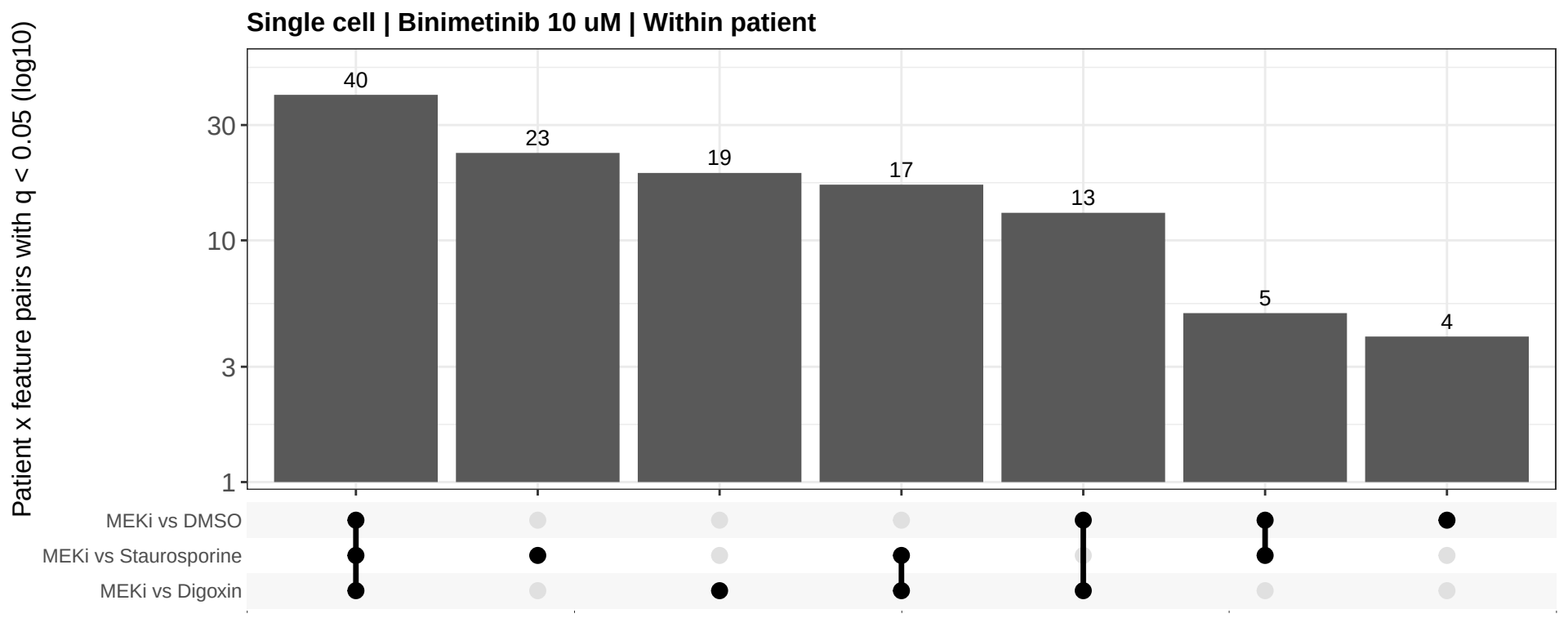

Single cell | Binimetinib 1 uM | Within tumor type
No significant features for: MEKi vs DMSO
Tumor type x feature pairs with q < 0.05 (log10)
129
3
100
10
1
MEKi vs Staurosporine
MEKi vs Digoxin

Single cell | Binimetinib 1 uM | Across tumor types (shared)

No features with q < 0.05

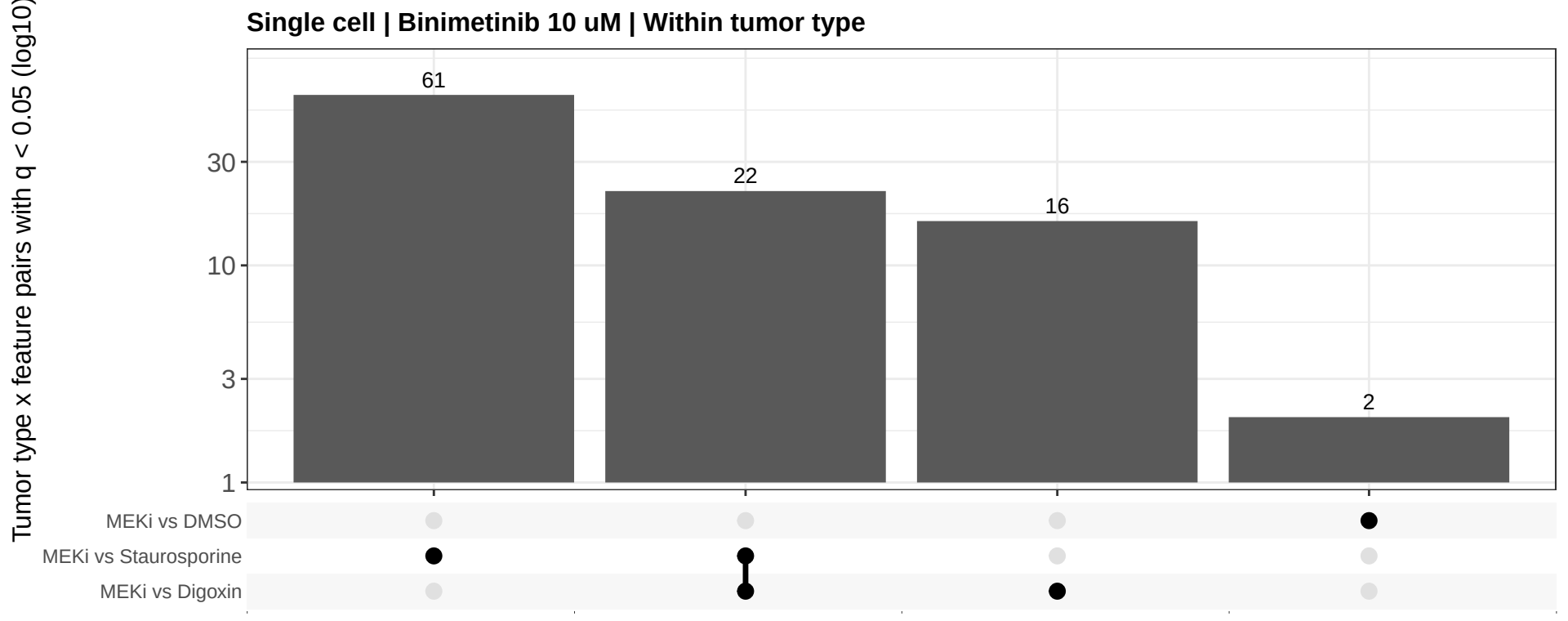
Single cell | Binimetinib 1 uM | Across tumor types (differ)

No features with q < 0.05


Single cell | Binimetinib 10 uM | Within patient
Patient x feature pairs with q < 0.05 (log10)
40
23
19
17
13
5
4
30
10
3
1
MEKi vs DMSO
MEKi vs Staurosporine
MEKi vs Digoxin

Single cell | Binimetinib 10 uM | Across patients

No features with q < 0.05

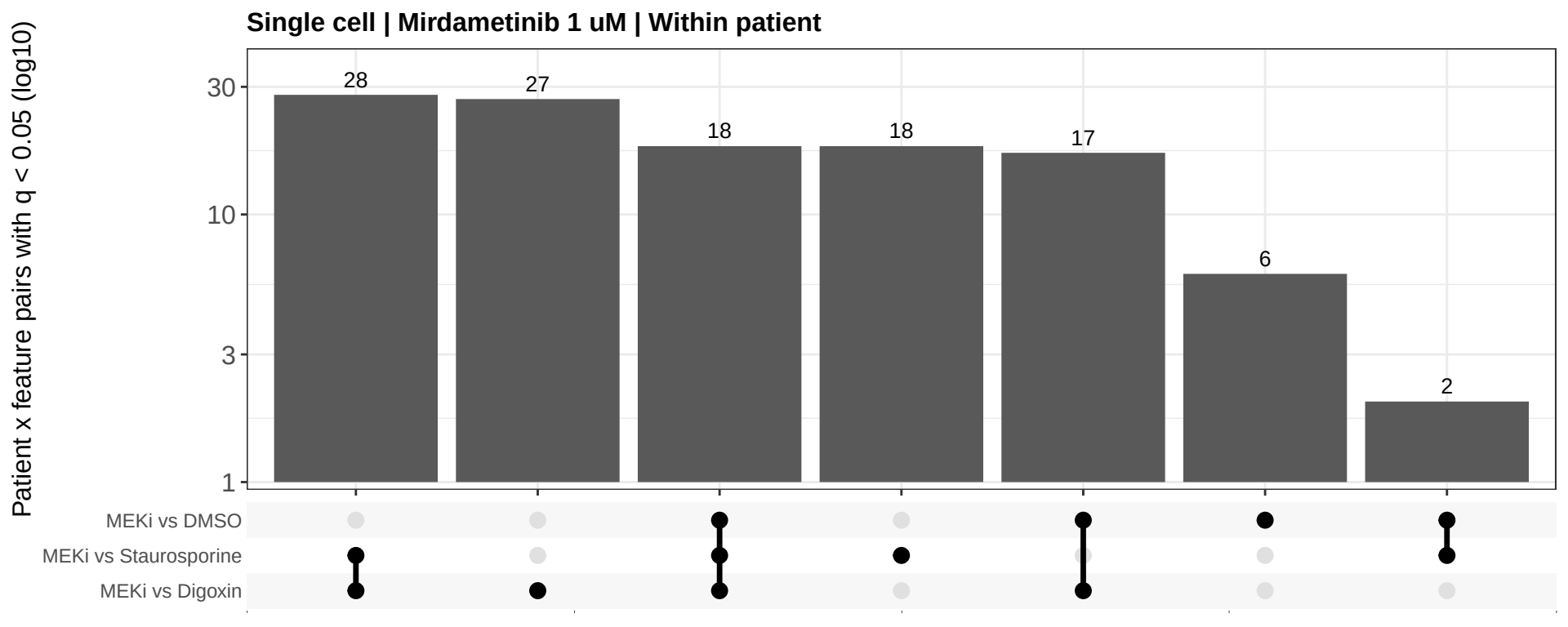

Single cell | Binimetinib 10 uM | Within tumor type
Tumor type x feature pairs with q < 0.05 (log10)
61
22
16
2
30
10
3
1
MEKi vs DMSO
MEKi vs Staurosporine
MEKi vs Digoxin

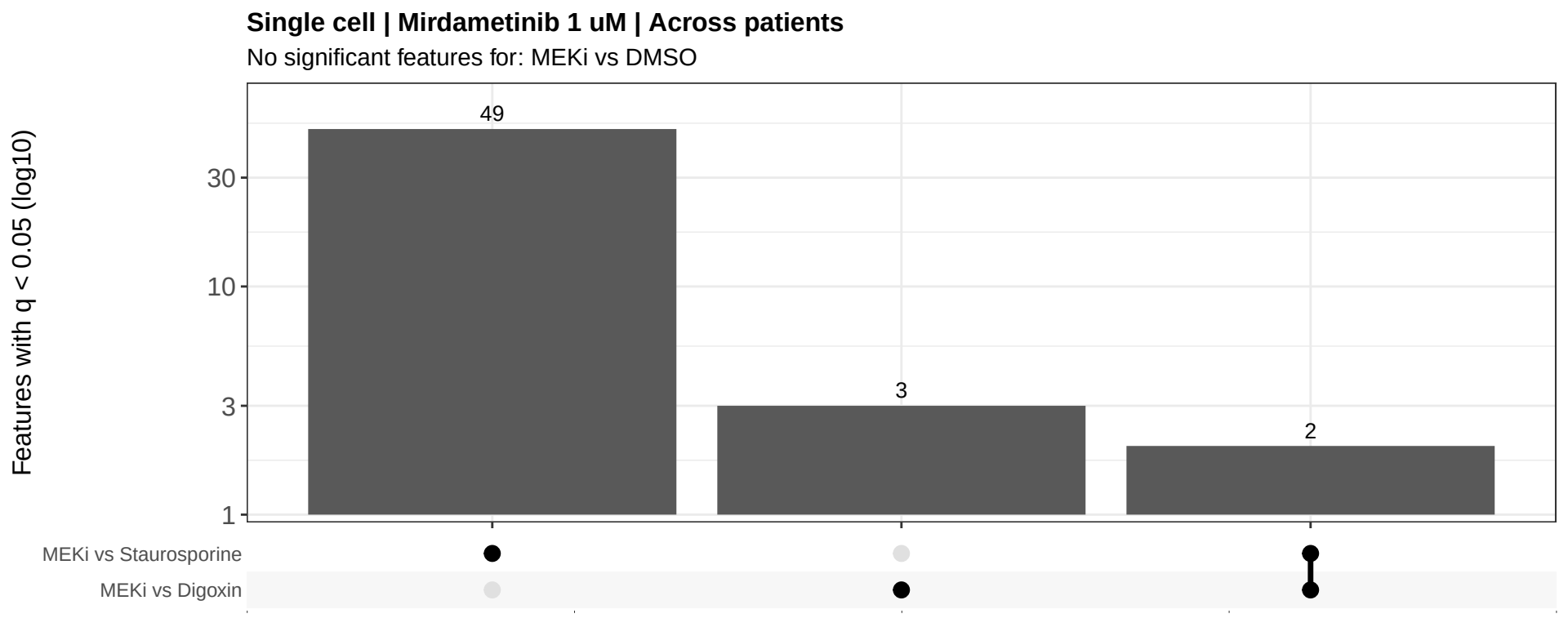
Single cell | Binimetinib 10 uM | Across tumor types (shared)

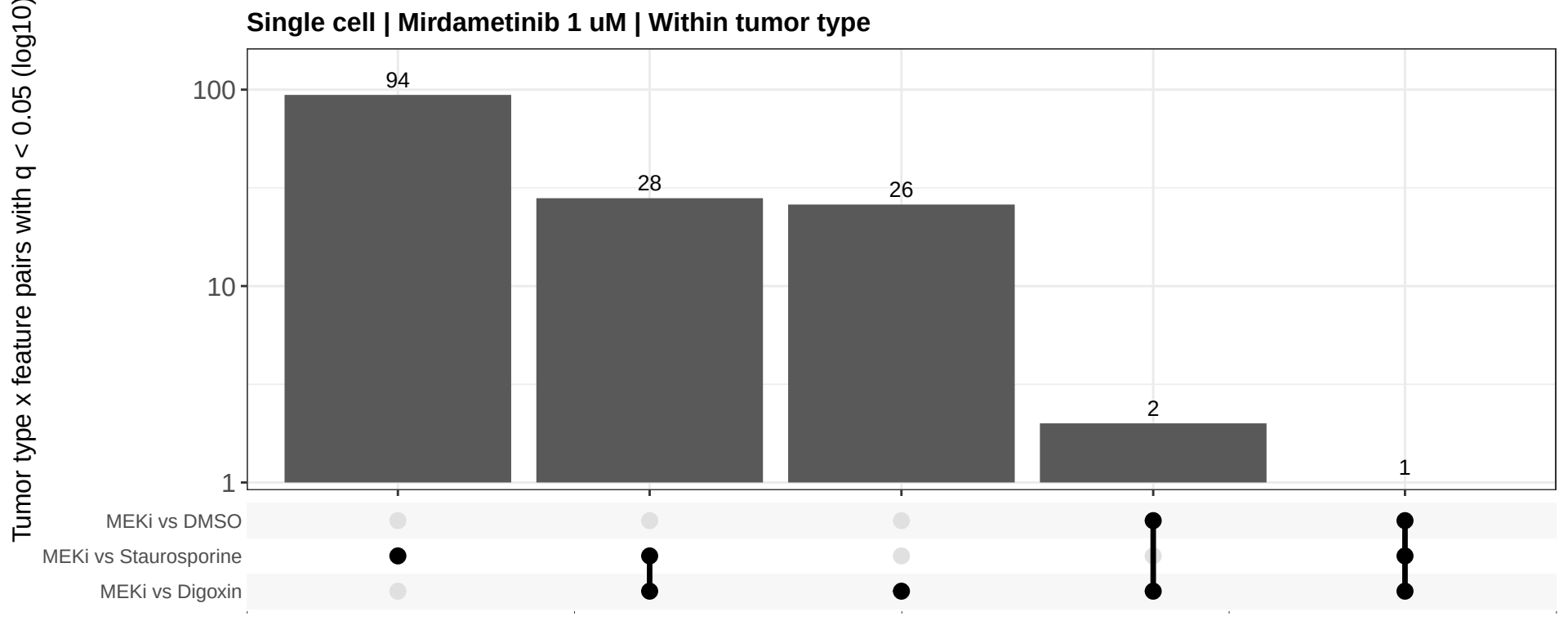
No features with q < 0.05

Single cell | Binimetinib 10 uM | Across tumor types (differ)

No features with q < 0.05


Single cell | Mirdametinib 1 uM | Within patient
Patient x feature pairs with q < 0.05 (log10)
30
10
3
1
28
27
18
18
17
6
2
MEKi vs DMSO
MEKi vs Staurosporine
MEKi vs Digoxin


Single cell | Mirdametinib 1 uM | Across patients
No significant features for: MEKi vs DMSO
Features with q < 0.05 (log10)
49
3
2
30
10
3
1
MEKi vs Staurosporine
MEKi vs Digoxin

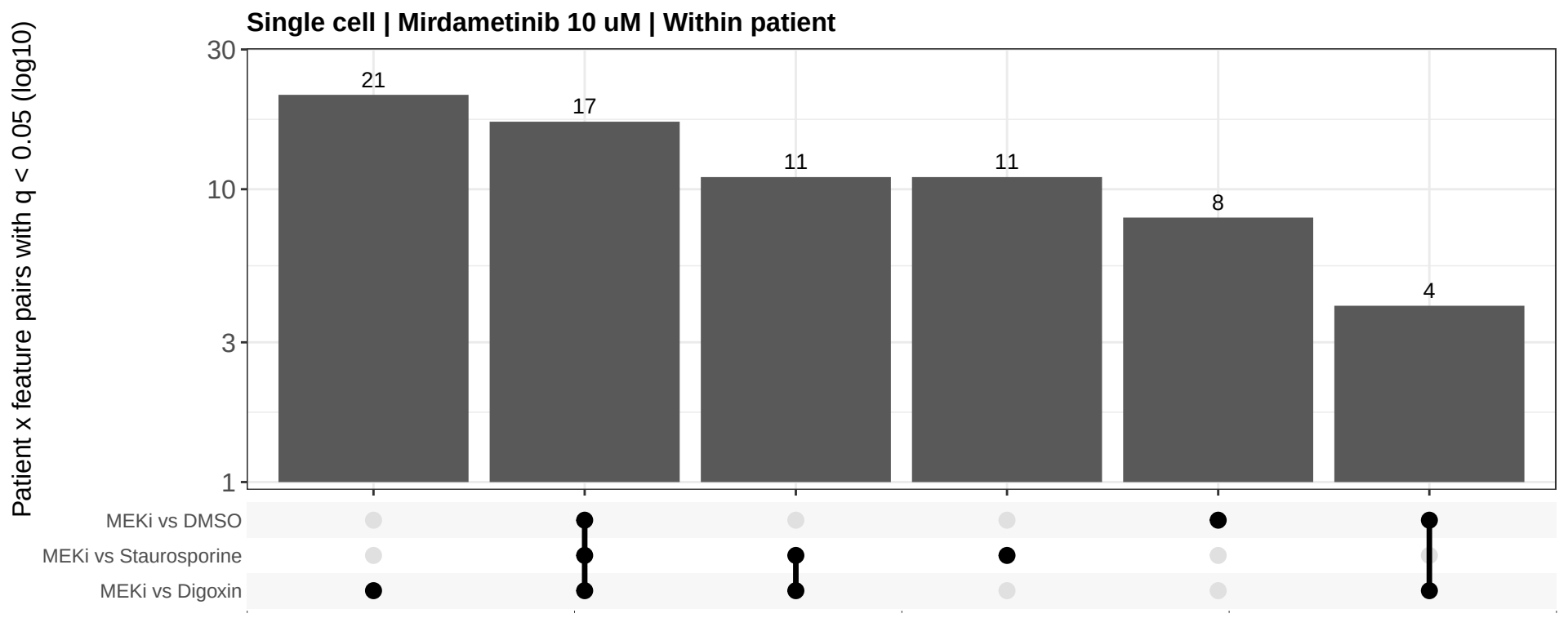

Single cell | Mirdametinib 1 uM | Within tumor type
Tumor type x feature pairs with q < 0.05 (log10)
100
10
1
94
28
26
2
1
MEKi vs DMSO
MEKi vs Staurosporine
MEKi vs Digoxin

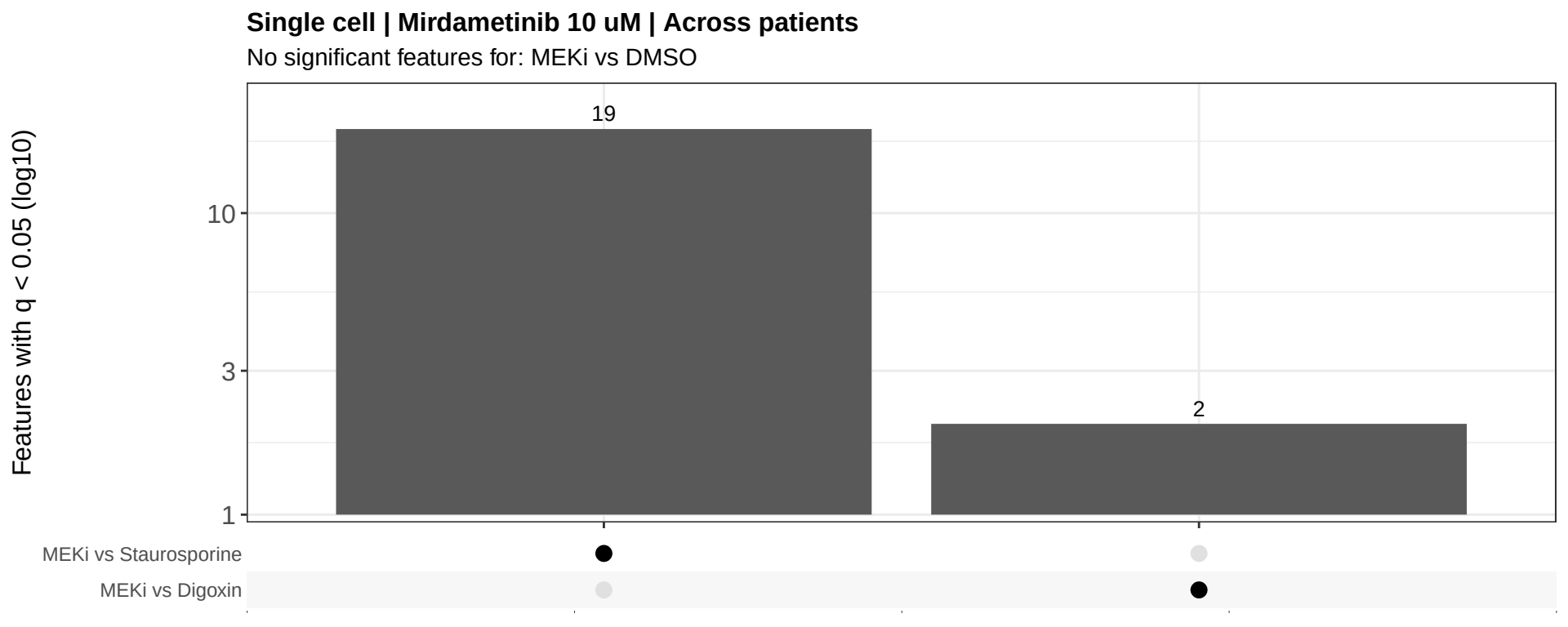
Single cell | Mirdametinib 1 uM | Across tumor types (shared)

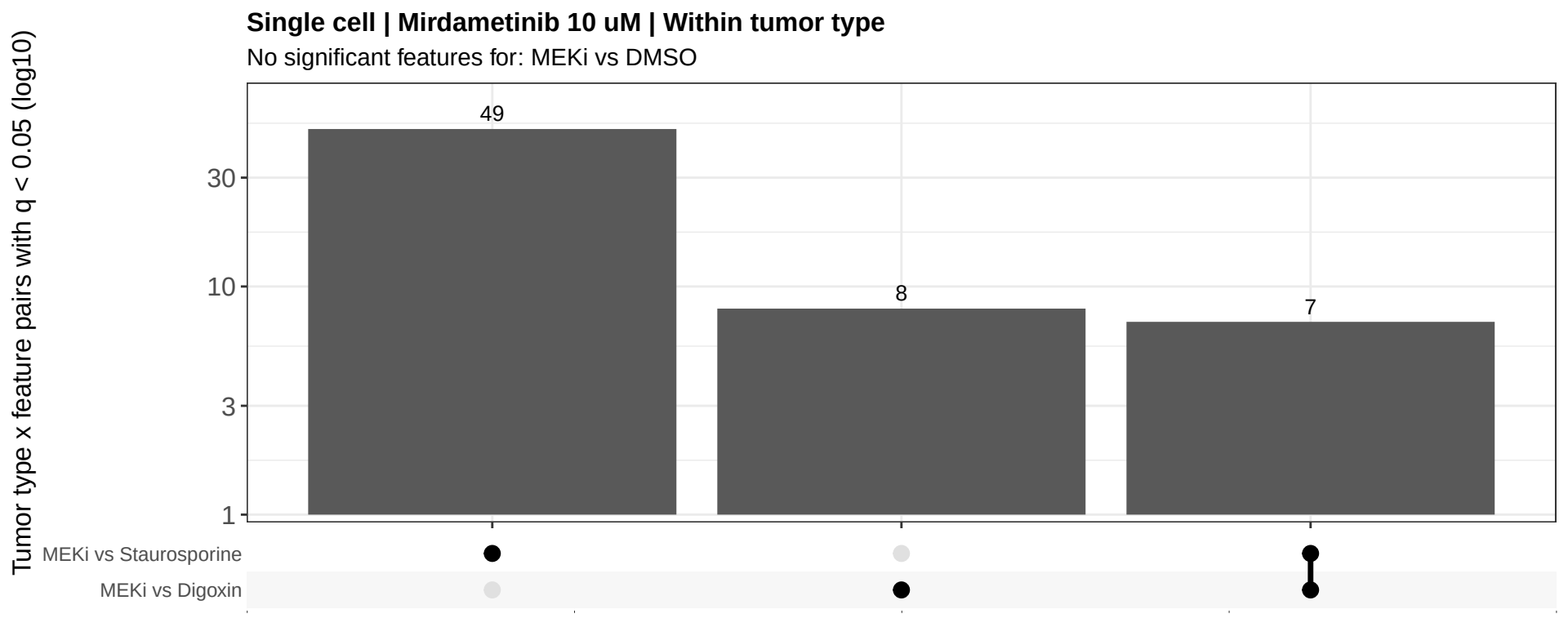
No features with q < 0.05

Single cell | Mirdametinib 1 uM | Across tumor types (differ)

No features with q < 0.05


Single cell | Mirdametinib 10 uM | Within patient
Patient x feature pairs with q < 0.05 (log10)
30
10
3
1
21
17
11
11
8
4
MEKi vs DMSO
MEKi vs Staurosporine
MEKi vs Digoxin


Single cell | Mirdametinib 10 uM | Across patients
No significant features for: MEKi vs DMSO
Features with q < 0.05 (log10)
19
2
10
3
1
MEKi vs Staurosporine
MEKi vs Digoxin

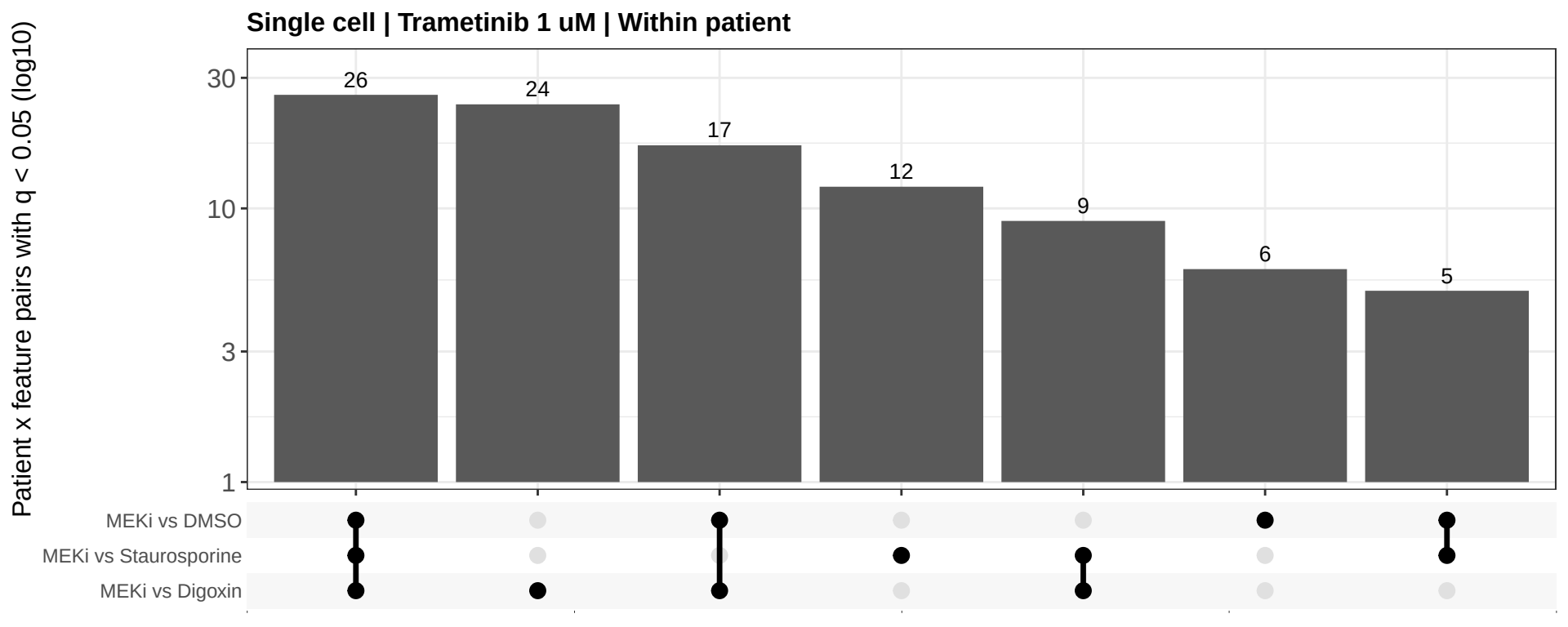

Single cell | Mirdametinib 10 uM | Within tumor type
No significant features for: MEKi vs DMSO
Tumor type x feature pairs with q < 0.05 (log10)
49
8
7
30
10
3
1
MEKi vs Staurosporine
MEKi vs Digoxin

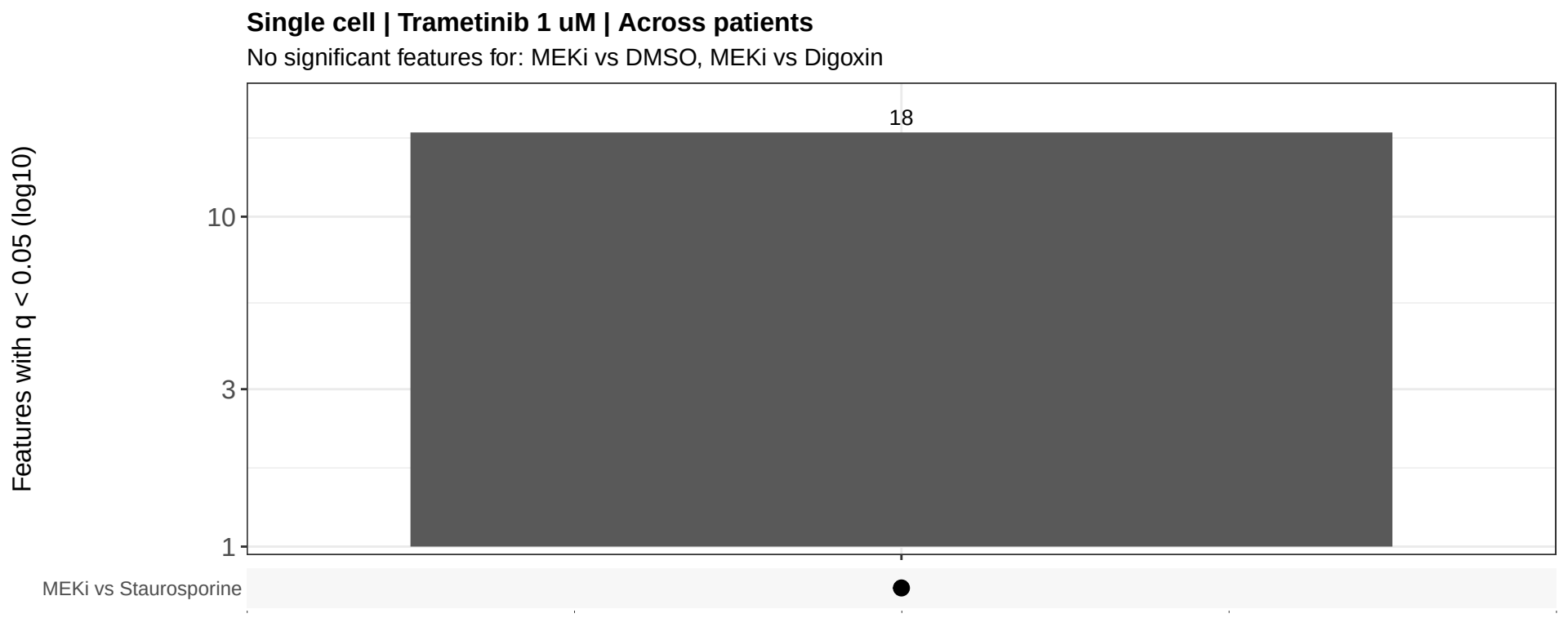
Single cell | Mirdametinib 10 uM | Across tumor types (shared)

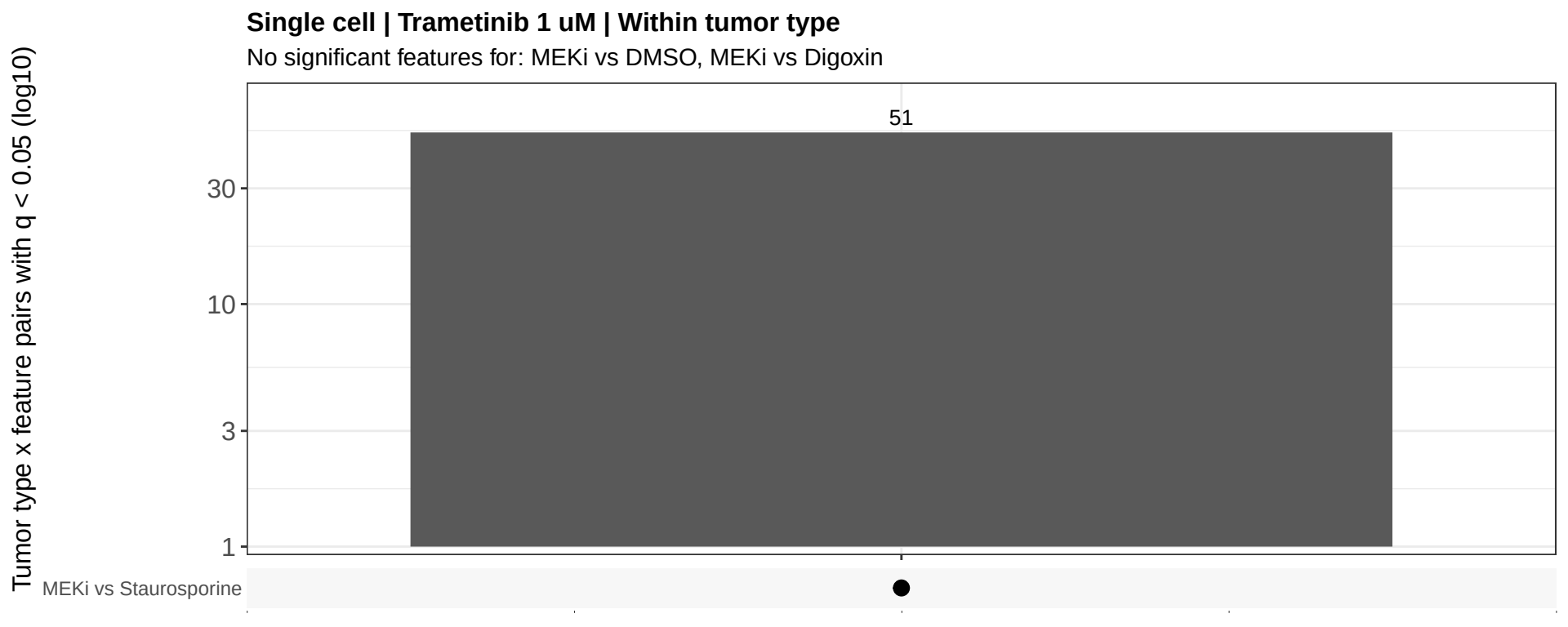
No features with q < 0.05

Single cell | Mirdametinib 10 uM | Across tumor types (differ)

No features with q < 0.05


Single cell | Trametinib 1 uM | Within patient
Patient x feature pairs with q < 0.05 (log10)
30
10
3
1
26
24
17
12
9
6
5
MEKi vs DMSO
MEKi vs Staurosporine
MEKi vs Digoxin


Single cell | Trametinib 1 uM | Across patients
No significant features for: MEKi vs DMSO, MEKi vs Digoxin
Features with q < 0.05 (log10)
18
10
3
1
MEKi vs Staurosporine

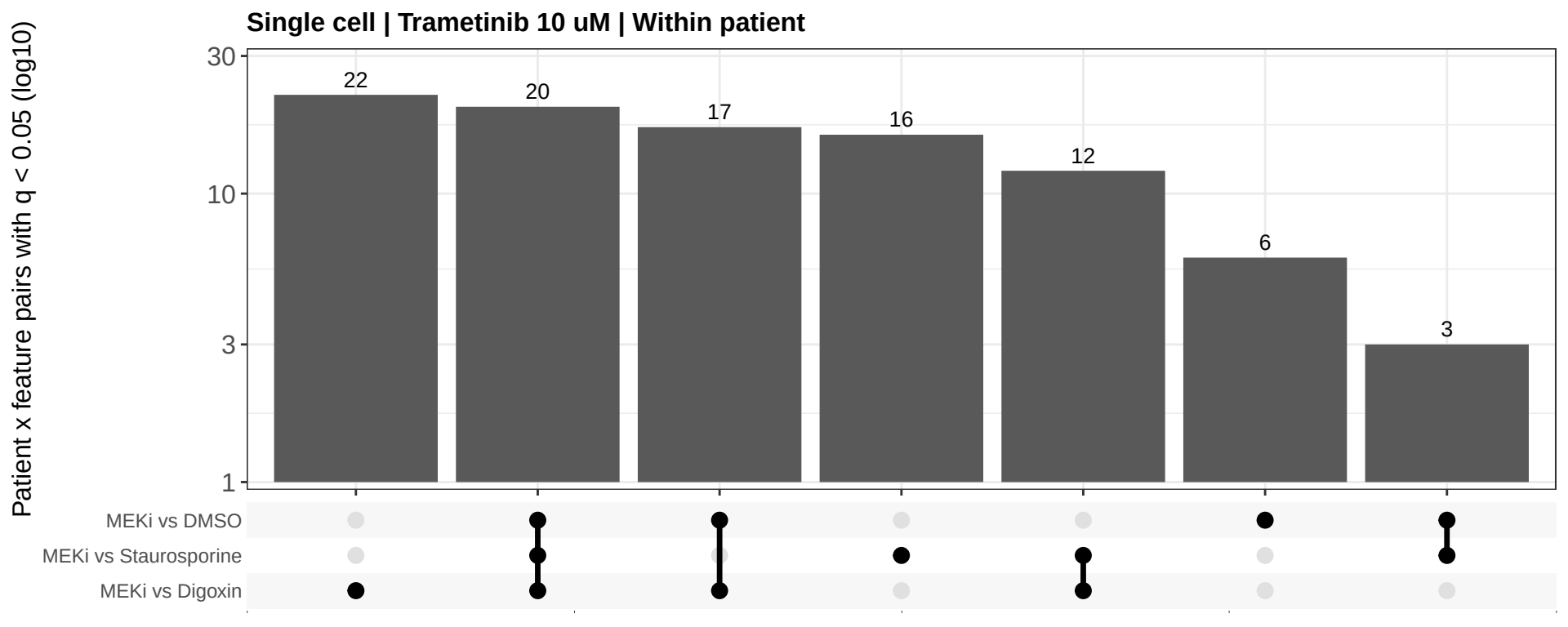

Single cell | Trametinib 1 uM | Within tumor type
No significant features for: MEKi vs DMSO, MEKi vs Digoxin
Tumor type x feature pairs with q < 0.05 (log10)
51
30
10
3
1
MEKi vs Staurosporine

Single cell | Trametinib 1 uM | Across tumor types (shared)

No features with q < 0.05

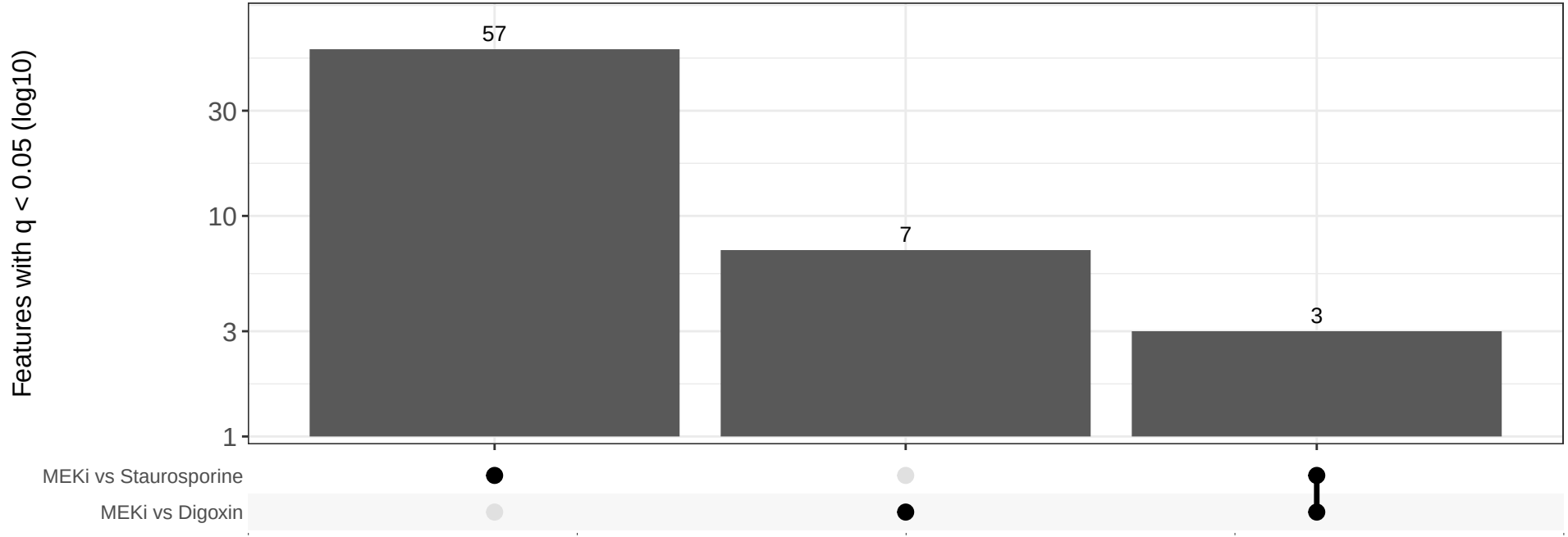
Single cell | Trametinib 1 uM | Across tumor types (differ)

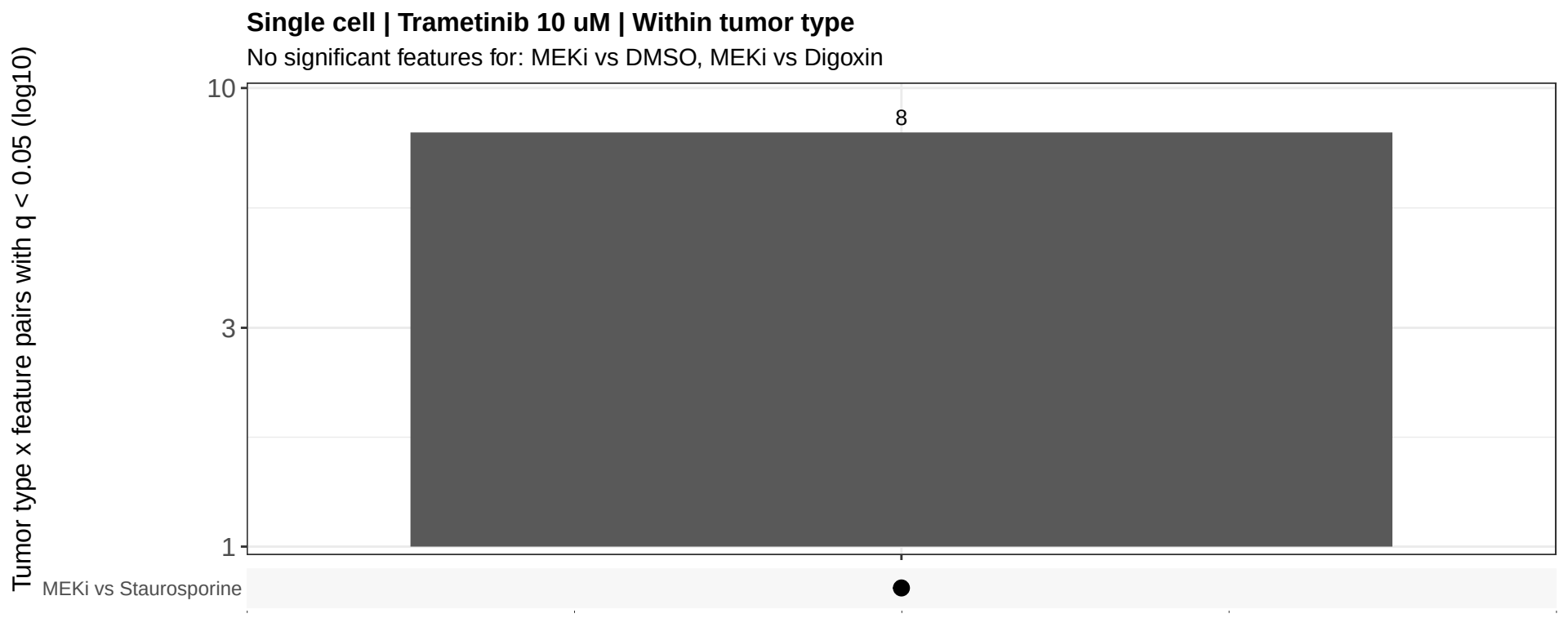
No features with q < 0.05


Single cell | Trametinib 10 uM | Within patient
Patient x feature pairs with q < 0.05 (log10)
30
10
3
1
22
20
17
16
12
6
3
MEKi vs DMSO
MEKi vs Staurosporine
MEKi vs Digoxin

Single cell | Trametinib 10 uM | Across patients

No significant features for: MEKi vs DMSO



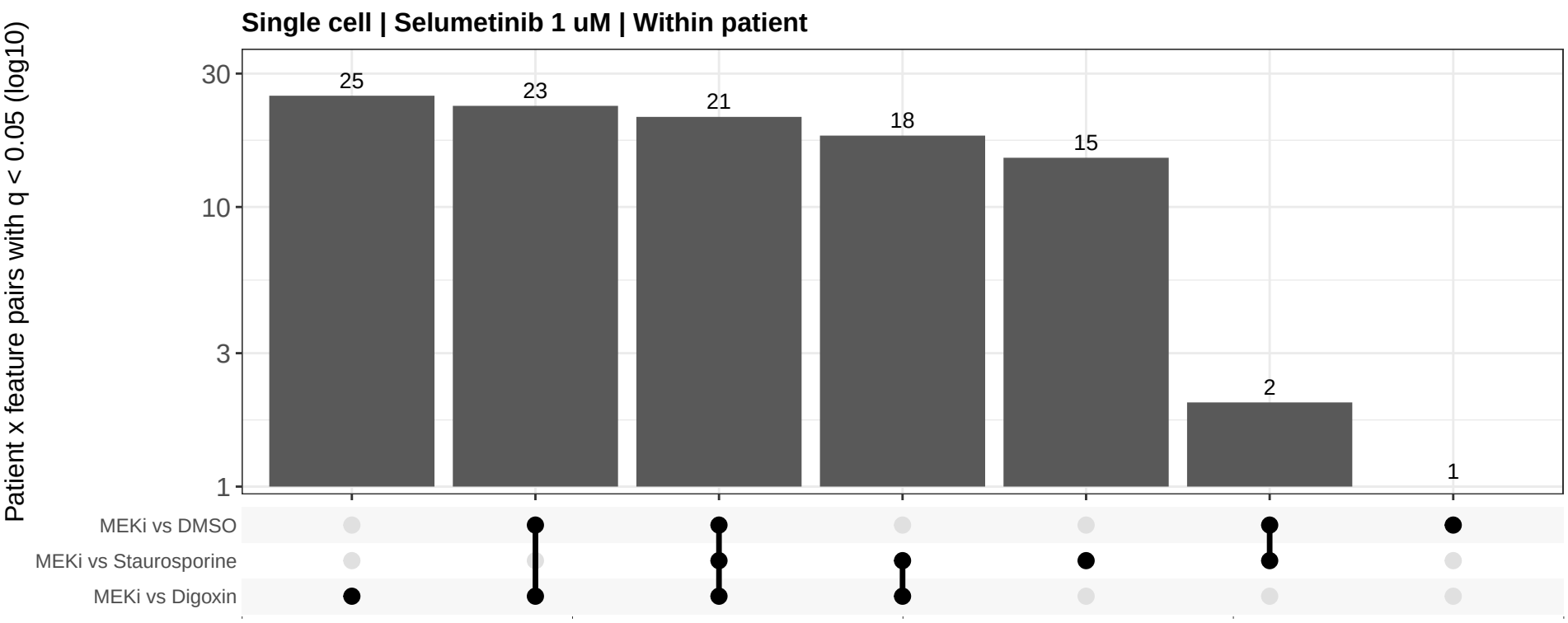

Single cell | Trametinib 10 uM | Within tumor type
No significant features for: MEKi vs DMSO, MEKi vs Digoxin
Tumor type x feature pairs with q < 0.05 (log10)
10
3
1
8
MEKi vs Staurosporine

Single cell | Trametinib 10 uM | Across tumor types (shared)

No features with q < 0.05

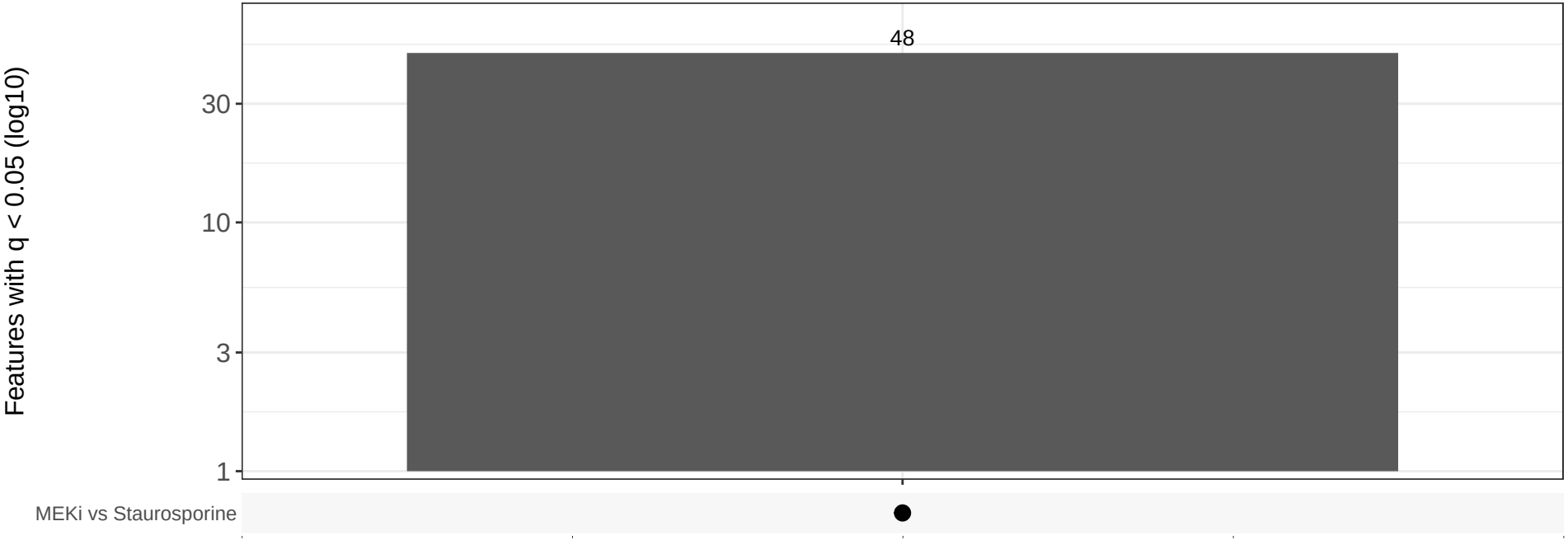
Single cell | Trametinib 10 uM | Across tumor types (differ)

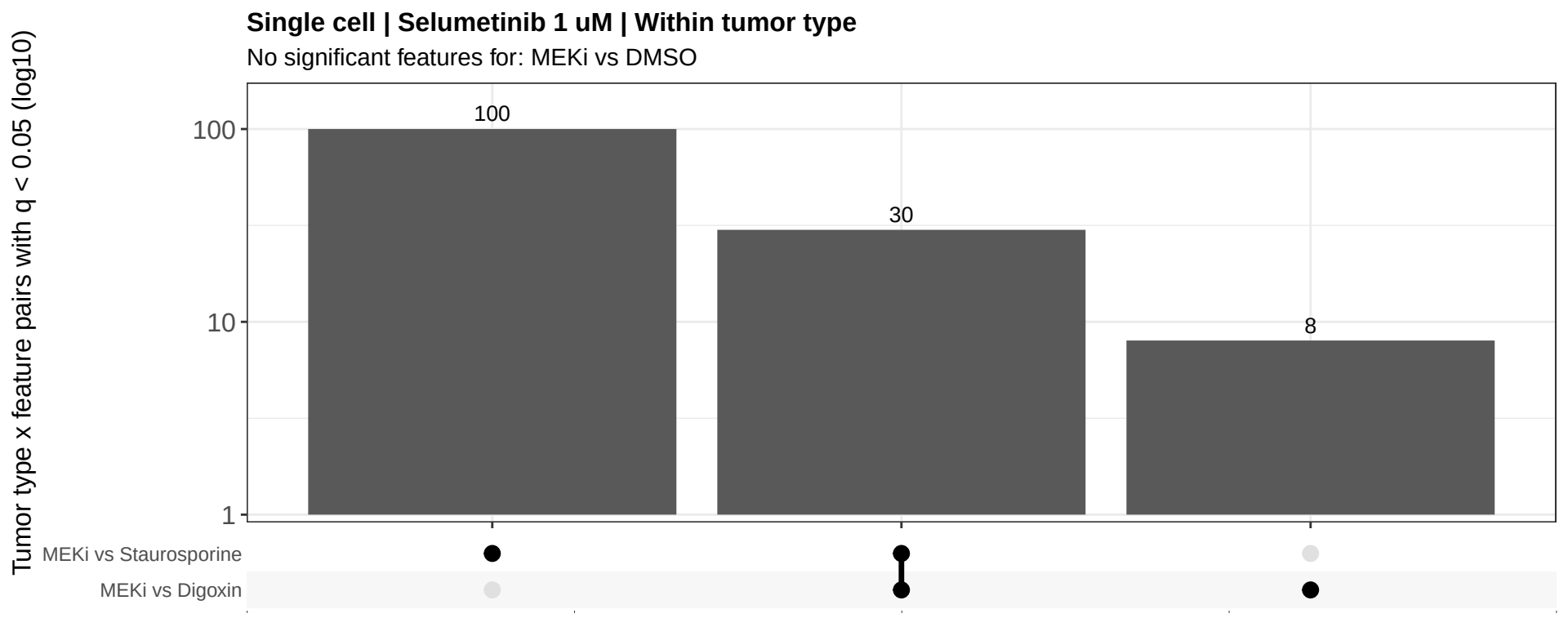
No features with q < 0.05


Single cell | Selumetinib 1 uM | Within patient
Patient x feature pairs with q < 0.05 (log10)
30
10
3
1
25
23
21
18
15
2
1
MEKi vs DMSO
MEKi vs Staurosporine
MEKi vs Digoxin

Single cell | Selumetinib 1 uM | Across patients

No significant features for: MEKi vs DMSO, MEKi vs Digoxin



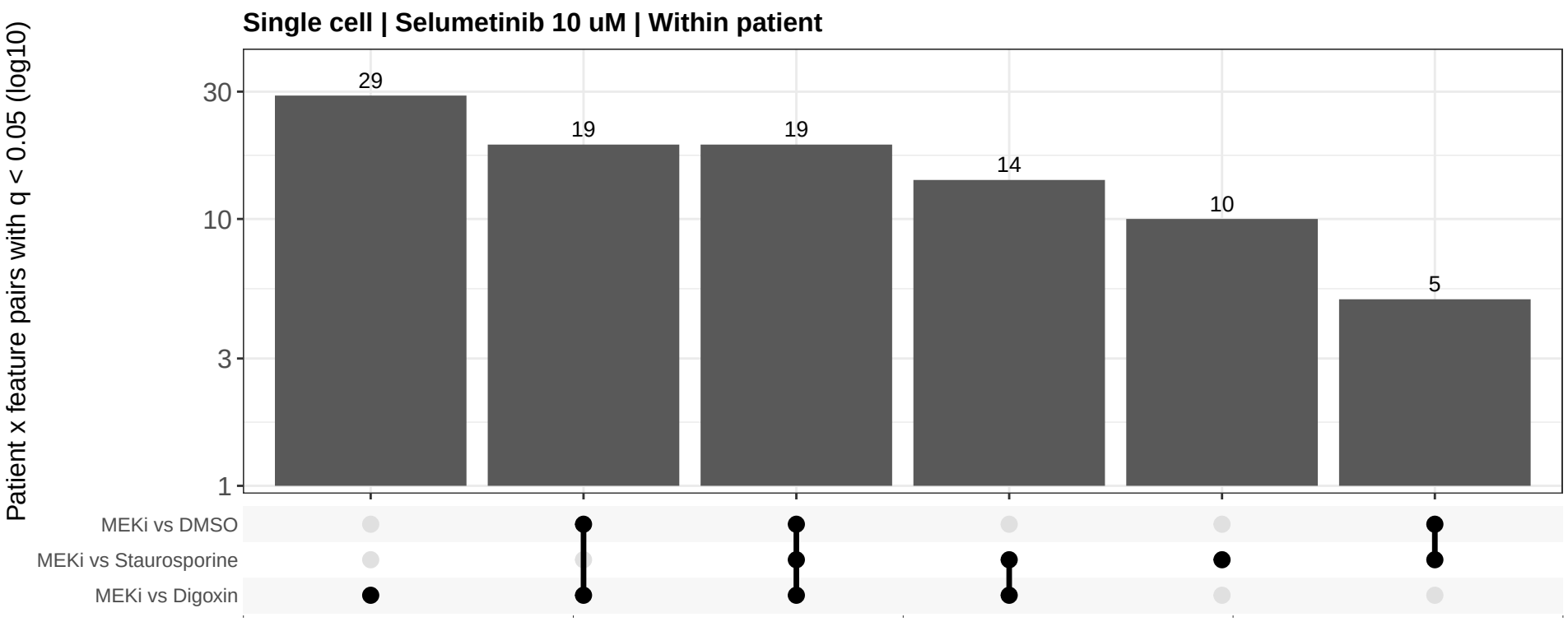

Single cell | Selumetinib 1 uM | Within tumor type
No significant features for: MEKi vs DMSO
Tumor type x feature pairs with q < 0.05 (log10)
100
10
1
100
30
8
MEKi vs Staurosporine
MEKi vs Digoxin

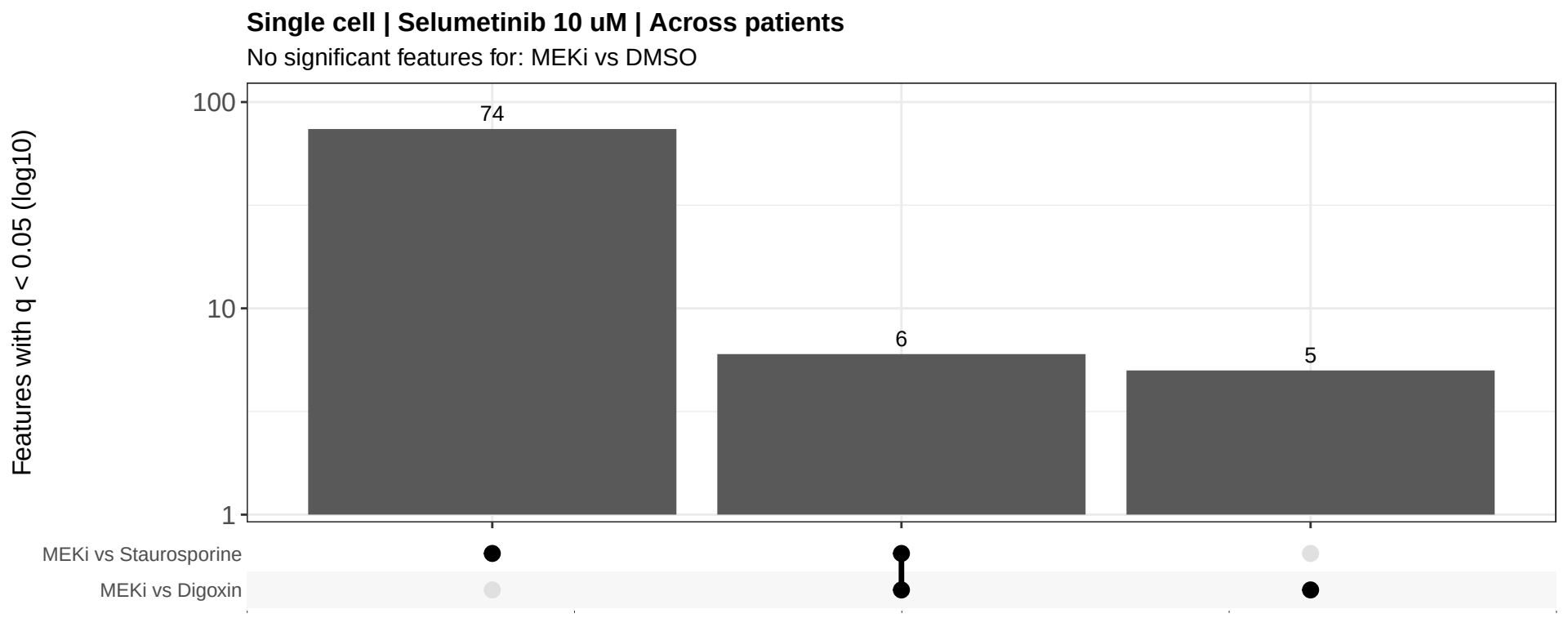
Single cell | Selumetinib 1 uM | Across tumor types (shared)

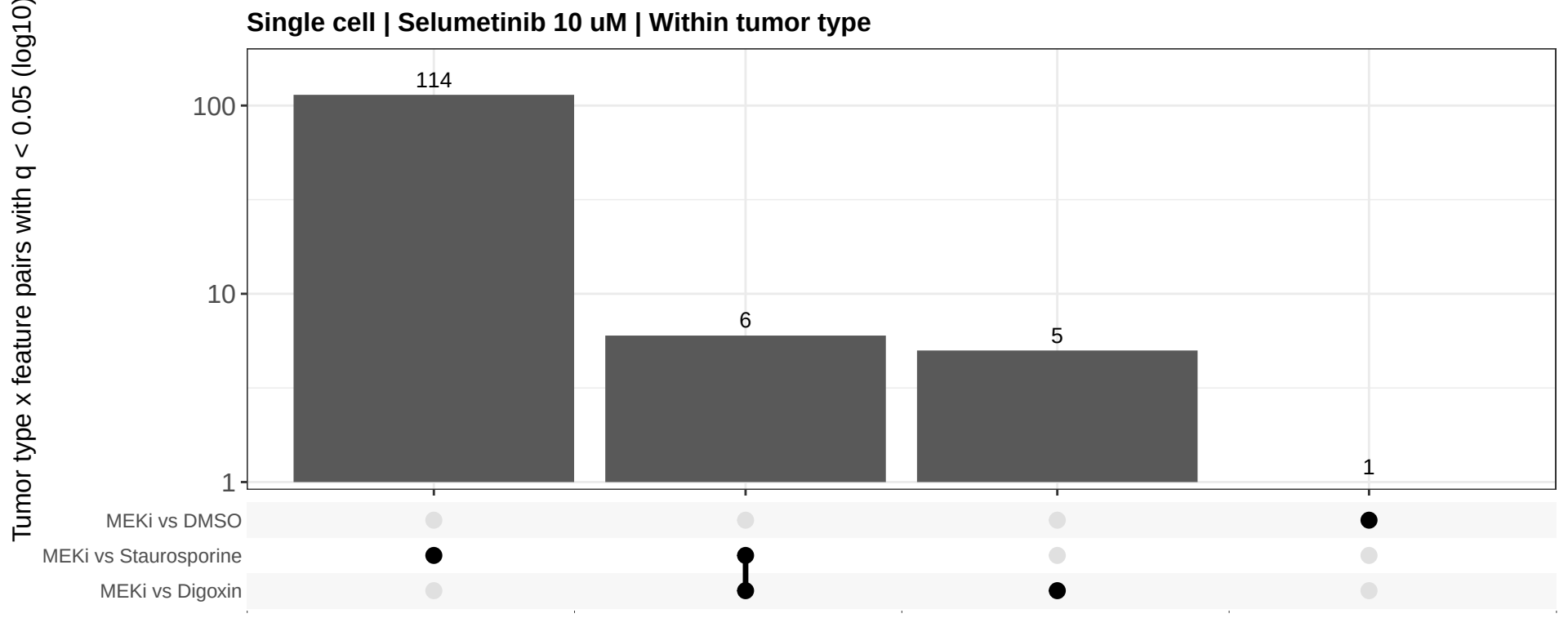
No features with q < 0.05

Single cell | Selumetinib 1 uM | Across tumor types (differ)

No features with q < 0.05


Single cell | Selumetinib 10 uM | Within patient
Patient x feature pairs with q < 0.05 (log10)
30
10
3
1
29
19
19
14
10
5
MEKi vs DMSO
MEKi vs Staurosporine
MEKi vs Digoxin


Single cell | Selumetinib 10 uM | Across patients
No significant features for: MEKi vs DMSO
Features with q < 0.05 (log10)
100
10
1
74
6
5
MEKi vs Staurosporine
MEKi vs Digoxin


Single cell | Selumetinib 10 uM | Within tumor type
Tumor type x feature pairs with q < 0.05 (log10)
100
10
1
114
6
5
1
MEKi vs DMSO
MEKi vs Staurosporine
MEKi vs Digoxin

Single cell | Selumetinib 10 uM | Across tumor types (shared)

No features with q < 0.05

Single cell | Selumetinib 10 uM | Across tumor types (differ)

No features with q < 0.05